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Methods

Calibration of Distributionally Robust Empirical Optimization Models

Jun-ya Gotoh,[a] Michael Jong Kim,[b] Andrew E. B. Lim[c,d]

[a] Department of Industrial and Systems Engineering, Chuo University, Tokyo 112-8551, Japan; [b] UBC Sauder School of Business, University of British Columbia, Vancouver, British Columbia V6T 1Z2, Canada; [c] Department of Analytics and Operations and Department of Finance, NUS Business School, National University of Singapore, Singapore 119245; [d] Institute for Operations Research and Analytics, National University of Singapore, Singapore 117602

Contact: jgoto@indsys.chuo-u.ac.jp, https://orcid.org/0000-0002-0097-7298 (JG); mike.kim@sauder.ubc.ca, https://orcid.org/0000-0002-1858-4142 (MJK); andrewlim@nus.edu.sg, https://orcid.org/0000-0001-7189-3406 (AEBL)

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Abstract. We study the out-of-sample properties of robust empirical optimization problems with smooth ϕ-divergence penalties and smooth concave objective functions, and we develop a theory for data-driven calibration of the nonnegative "robustness parameter" δ that controls the size of the deviations from the nominal model. Building on the intuition that robust optimization reduces the sensitivity of the expected reward to errors in the model by controlling the spread of the reward distribution, we show that the first-order benefit of "little bit of robustness" (i.e., δ small, positive) is a significant reduction in the variance of the out-of-sample reward, whereas the corresponding impact on the mean is almost an order of magnitude smaller. One implication is that substantial variance (sensitivity) reduction is possible at little cost if the robustness parameter is properly calibrated. To this end, we introduce the notion of a robust mean-variance frontier to select the robustness parameter and show that it can be approximated using resampling methods such as the bootstrap. Our examples show that robust solutions resulting from "open-loop" calibration methods (e.g., selecting a 90% confidence level regardless of the data and objective function) can be very conservative out of sample, whereas those corresponding to the robustness parameter that optimizes an estimate of the out-of-sample expected reward (e.g., via the bootstrap) with no regard for the variance are often insufficiently robust.

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Keywords: distributionally robust optimization • calibration • worst-case sensitivity • variance reduction

1. Introduction

Empirical optimization (or sample average approximation (SAA)) is an approach for expected value optimization using the empirical distribution as an alternative model to the unknown true distribution. Misspecification can occur because of incorrect modeling assumptions or estimation uncertainty, and decisions that are made on the basis of an incorrect model can perform poorly out of sample if misspecification is ignored. Distributionally robust optimization (DRO) accounts for misspecification in the in-sample problem by optimizing against worst-case perturbations from the "nominal model."

DRO models are typically parameterized by an "ambiguity parameter" δ that controls the size of the deviations from the nominal model in the worst-case problem. The ambiguity parameter may appear as the confidence level of an uncertainty set or a penalty parameter that multiplies some measure of deviation between alternative probability distributions and the nominal in the worst-case objective. The parameter δ defines a family of worst-case solutions $\{x_n(\delta) : \delta \geq 0\}$, where $\delta = 0$ is empirical/sample average optimization (i.e., no robustness) with solutions becoming increasingly conservative as δ increases. (Here, n denotes the size of the historical data set used to

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construct the robust optimization problem.) The choice of δ clearly determines the out-of-sample performance of the robust solution. The goal of this paper is to study the out-of-sample properties of DRO and to develop a theory for data-driven calibration of the "robustness parameter" for worst-case maximization problems with concave reward functions.

1.1. Summary of Contributions

This paper makes the following contributions.

1. *DRO is a multiobjective problem.* In the context of our model, we show that the sensitivity of the expected reward to worst-case deviations from the nominal is equal to the variance of the reward. Building on this insight and the results from Gotoh et al. (2018), we show that DRO is a multiobjective problem where the goal is to maximize the *expected reward* while controlling the *worst-case sensitivity* (variance) to misspecification. The trade-off between mean and sensitivity (variance) is controlled by the robustness parameter δ, which suggests that both objectives should be considered when selecting δ.

2. *Out-of-sample properties of DRO and worst-case sensitivity reduction.* We show that solutions of SAA and DRO are asymptotically normal (n large), that a little robustness (δ small) always reduces the sensitivity (variance) of the out-of-sample reward, and that sensitivity (variance) reduction is almost an order of magnitude larger than the impact on the mean when δ is small. We also show that the out-of-sample expected reward under the worst-case solution can sometimes be larger than that of SAA, though this improvement is typically small relative to sensitivity/variance reduction (see Bertsimas et al. (2013), Kim and Lim (2014), and Lorca and Sun (2014) for empirical observations of this phenomenon, and see Anderson and Philpott (2019) for the analysis in a one-dimensional quadratic setting).

3. *Calibration of the robustness parameter.* The problem of calibrating δ in DRO is analogous to the selection of the free parameters in various machine learning algorithms (e.g., the penalty parameter in LASSO, ridge regression, support vector machines), which are commonly tuned by optimizing an estimate of out-of-sample loss via cross-validation, bootstrap, or some other resampling approach. Although δ could be selected by optimizing an estimate of the out-of-sample expected reward, this ignores the multiobjective character of DRO, which suggests that both the mean and variance should be taken into account. We show how the out-of-sample robust mean-variance frontier can be constructed using resampling techniques. Consistent with our theory, our examples show that substantial variance reduction is possible with minimal impact on the expected reward if δ is chosen appropriately. The values of δ obtained from our approach are typically much smaller than those implied by standard confidence levels (e.g., 90%, 95%) advocated in the literature and larger than those obtained by maximizing only the mean.

1.2. Other Related Literature

Several papers (Duchi et al. 2021, Lam 2016, Namkoong and Duchi 2017, Gotoh et al. 2018) discuss DRO from the perspective of empirical optimization with the variance of the reward as a regularizer. Most related is Gotoh et al. (2018), which studies the connection between DRO with (sufficiently smooth) ϕ-divergence penalties and mean-variance problems when the robustness parameter is small. The most important difference is that Gotoh et al. (2018) studies the in-sample DRO problem, whereas this paper is an out-of-sample analysis. Specifically, we characterize the impact of data variability on the DRO solution and the mean and variance of the out-of-sample reward, which forms the basis of resampling-based calibration methods for selecting the robustness parameter δ. Our analysis also reveals the mechanism by which the out-of-sample expected reward under DRO solutions can exceed that of SAA, a phenomenon that has been observed empirically in a number of papers but not fully explained.

A common theme in Duchi et al. (2021), Gotoh et al. (2018), Lam (2016), Namkoong and Duchi (2017), and the present paper is the relationship between DRO and variance regularization of SAA, and it is natural to ask *why* such a relationship even exists. In this regard, we show that the in-sample variance of the reward is equal to the sensitivity of the mean under worst-case deviations from the nominal, so optimizing against the worst case in DRO is almost the same as maximizing the expected reward while controlling worst-case sensitivity to model misspecification (variance). This formalizes discussion in Gotoh et al. (2018) and provides insight into why DRO and variance regularization of SAA are related. Variance regularization is also proposed in Ban et al. (2018) as a way to reduce the impact of data variability of the out-of-sample performance of the solution of mean Conditional Value-at-Risk (CVaR) portfolio choice problems, though connections to worst-case optimization are not discussed; see also Gotoh et al. (2018) for the relationship between worst-case CVaR and variance regularization.

Also related is the paper by Namkoong and Duchi (2017), which develops finite sample probabilistic guarantees for the out-of-sample expected reward generated by robust solutions. One important difference is that Namkoong and Duchi (2017) studies the out-of-sample expected reward, whereas the focus of our paper is the variance/sensitivity reduction properties of DRO and the implications for calibration. Additionally, although the probabilistic guarantees in

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Namkoong and Duchi (2017) formalize the relationship between data size, model complexity, robustness, and out-of-sample expected reward, these results depend on quantities that are difficult to compute (e.g., the covering number or the Vapnik–Chervonenkis dimension), require a bounded objective function, and come with the usual concerns that performance bounds of this nature are loose (Abu-Mostafa et al. 2012). Note, too, that Namkoong and Duchi (2017) is a finite sample analysis, whereas ours is asymptotic (large sample size), though calibration experiments for small data sets produce results that are consistent with our large sample theory. We also mention the paper by Duchi et al. (2021), which provides confidence intervals for the optimal objective value and shows consistency of solutions using Owen's empirical likelihood theory, and that of Lam (2016), which studies the sensitivity of estimates of the expected value of random variables to worst-case perturbations of a simulation model.

Connections between robustness and regularization are studied in Bertsimas and Copenhaver (2018), Blanchet et al. (2017), Gao et al. (2017), and Xu et al. (2010), where various worst-case regression and classification problems are shown to be equivalent to the regularized regression and classification problems (ridge, Lasso, etc.). Compared with this paper and also Namkoong and Duchi (2017), Duchi et al. (2016), Gotoh et al. (2018), and Lam (2016), one important difference is that the regularizers in Bertsimas and Copenhaver (2018), Blanchet et al. (2017), Gao et al. (2017), and Xu et al. (2010) act on the solution, whereas the sensitivity/variance penalty in this paper acts on the reward. The equivalence with solution regularization in Bertsimas and Copenhaver (2018), Blanchet et al. (2017), Gao et al. (2017), and Xu et al. (2010) implies that worst-case regression and classification reduces the variance of the *solution* (regression estimate) by shrinking it toward the origin. It turns out, however, that this is not a general property of DRO but is driven by the objective function associated with regression and classification. Indeed, we show in Section 3 that there are well-behaved concave reward functions where the impact of DRO on the SAA solution is opposite to that of a solution regularizer, biasing it away from the origin and increasing its variance.

We comment briefly on various approaches to calibrating DRO models in the literature. One approach is to use *high confidence uncertainty sets* that include the true data generating model with high probability, where the confidence level (typically 90%, 95%, or 99%) is a primitive of the model (Delage and Ye 2010, Ben-Tal et al. 2013, Duchi et al. 2021, Lam 2016, Bertsimas et al. 2018b). This has been refined by Bertsimas et al. (2018a), who develop methods for finding the smallest such uncertainty set, though the confidence level remains a primitive of the model. One concern with this approach is that it is "open loop," with confidence levels chosen independent of the data and the objective function. There is, however, no reason why these particular confidence levels should have anything to do with good out-of-sample performance.

A second approach selects the robustness parameter δ by *optimizing an estimate of out-of-sample expected reward* using a resampling procedure such as a bootstrap or cross-validation. Although δ now depends on the data and the objective function (it is no longer "open loop"), it ignores the multiobjective character of DRO. Indeed, our results show that sensitivity/variance reduction is the first-order benefit in DRO, so optimizing just the mean corresponds to selecting δ on the basis of (almost) a second-order effect. Although optimizing the mean can produce a solution that has a higher out-of-sample expected reward than empirical optimization in certain applications, there are other applications where, by the same criteria, it is optimal not to be robust (i.e., $\delta = 0$). In either case, there are typically larger values of δ where further reduction in the sensitivity/variance is possible with negligible impact on the mean. In contrast to the "high-confidence" approach where classical confidence levels of 90%, 95%, and 99% produce overly conservative solutions, this approach is likely to be insufficiently robust.

Finally, in the *satisfycing approach* of Brown et al. (2012), the decision maker specifies a target level T and finds the "most robust" decision that achieves this target under the worst-case *in-sample* alternative model. That is, he or she chooses the largest robustness parameter δ under which the worst-case expected reward exceeds the target T, so the target T is a primitive of the problem, and the confidence level δ is optimized.

One surprising property of DRO is that the out-of-sample expected reward can sometimes exceed that of the SAA solution, even though it is a worst-case approach. This has been observed empirically (Bertsimas et al. 2013, Kim and Lim 2014, Lorca and Sun 2014) and is studied in Anderson and Philpott (2019) for a quadratic reward and one-dimensional decision variable. We show that the difference between the out-of-sample expected reward under its "true" maximizer and the SAA decision is nothing but the "gap" in Jensen's inequality as a result of the variability of the solution (see Figure 2) and that the improvement in the out-of-sample expected reward under DRO, should it occur, is precisely the reduction in this gap under the DRO solution. Our results show that the change in the expected reward is likely to be small

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relative to the reduction in worst-case sensitivity, which is the primary benefit of the DRO model.

Finally, we mention the paper by Wu et al. (2017), which studies the asymptotic properties of stochastic optimization problems with risk-sensitive objectives.

1.3. Notation

Given any function f defined over the positive reals, we write $f(\delta) = o(\delta)$ (as $\delta \to 0$) provided $\lim_{\delta\to 0} f(\delta)/\delta = 0$, and we write $f(\delta) = O(\delta)$ (as $\delta \to 0$) provided there exists $M, \delta_0 > 0$ such that $f(\delta) \le M\delta$ for all $\delta \le \delta_0$. Given two stochastic processes $\{X_n\}$ and $\{Y_n\}$ defined on a common probability space, we write $X_n = o_P(Y_n)$ provided for any $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}(|X_n/Y_n| > \epsilon) = 0$.

1.4. Outline of the Remainder of the Paper

We introduce the in-sample DRO model and the notion of worst-case sensitivity in Section 2. We also show that worst-case sensitivity is equal to the variance of the reward for all sufficiently smooth ϕ-divergence and that our DRO problem is equivalent to a multiobjective mean-sensitivity problem when the robustness parameter is small. We characterize the statistical properties of the in-sample solutions in Section 3. We also show that it is possible for DRO to increase the variance of the solution relative to empirical optimization. We characterize the mean and variance of the associated out-of-sample rewards for empirical optimization ($\delta = 0$) in Section 4 and robust optimization ($\delta > 0$) in Section 5. In the case of DRO, we show that out-of-sample variance reduction is almost an order of magnitude larger than any change in the mean when the robustness parameter is small. We introduce our calibration method, which is based on resampled estimates of the mean and variance, in Section 6, and we apply this to three examples in Section 7. We conclude in Section 8.

2. Robust Empirical Optimization

Let $f(x, Y)$ be a real-valued reward function, where $x \in \mathbb{R}^d$ is the decision variable and $Y \in \mathbb{R}^l$ a random vector with distribution $\mathbb{P}$. If the distribution $\mathbb{P}$ of Y is known, we can solve

$$\max_x \mathbb{E}_{\mathbb{P}}[f(x, Y)], \tag{1}$$

(assumptions on f will be given later). In many situations, the distribution $\mathbb{P}$ is not known. Instead, we have historical data $Y_1, \ldots, Y_n$ generated independent and identically distributed (i.i.d.) from $\mathbb{P}$, and we would like to use this data set to make a decision that performs well out of sample.

When the distribution $\mathbb{P}$ of Y is not known, it is natural to optimize a sample average approximation of the objective

$$x_n(0) := \arg\max_x \left\{ \mathbb{E}_{\mathbb{P}_n}[f(x, Y)] \equiv \frac{1}{n}\sum_{i=1}^n f(x, Y_i) \right\}, \tag{2}$$

where $\mathbb{P}_n$ is the empirical distribution associated with $Y_1, \cdots, Y_n$. In many situations, however, $x_n(0)$ performs poorly out of sample because of differences between the empirical estimate $\mathbb{P}_n$ and the actual population distribution $\mathbb{P}$.

One approach to this problem is to optimize worst-case versions of the empirical optimization problem (2). Specifically, let $\delta > 0$ be a positive constant, and let

$$x_n(\delta) := \arg\max_x \left\{ \min_{\mathbb{Q}} \left\{ \mathbb{E}_{\mathbb{Q}}[f(x, Y)] + \frac{1}{\delta}\mathcal{H}_\phi(\mathbb{Q}\,|\,\mathbb{P}_n) \right\} \right\} \tag{3}$$

be the solution of the DRO problem where

$$\mathcal{H}_\phi(\mathbb{Q}\,|\,\mathbb{P}_n) := \begin{cases} \sum_{i:p_i^n>0} p_i^n \phi\left(\frac{q_i}{p_i^n}\right) & \sum_{i:p_i^n>0} q_i = 1, q_i \ge 0, \\ +\infty & \text{otherwise,} \end{cases} \tag{4}$$

is the ϕ-divergence of $\mathbb{Q} = [q_1, \cdots, q_n]$ relative to $\mathbb{P}_n = [p_1^n, \cdots, p_n^n]$. Although there are important examples of ϕ-divergence that are not smooth, we assume throughout this paper that ϕ is a convex function such that

$$\operatorname{dom}\phi \subset [0, \infty),\ \phi(1) = 0,\ \phi'(1) = 0, \text{ and } \phi''(1) > 0 \tag{5}$$

(additional assumptions will be stated as required). The ϕ-divergence $\mathcal{H}_\phi(\mathbb{Q}\,|\,\mathbb{P}_n)$ measures the deviation of the alternative distribution $\mathbb{Q}$ from the nominal $\mathbb{P}_n$, and the worst-case model (3) accounts for errors in the empirical distribution $\mathbb{P}_n$ by optimizing against worst-case perturbations. The size of the perturbations in (3) is controlled by the robustness parameter δ, which determines the penalty on the adversary for deviating from the nominal. Note that $\delta = 0$ gives the empirical model (2), and solutions become "more conservative" as δ increases.

It is shown in Gotoh et al. (2018) that if $\phi(z)$ is sufficiently smooth, the worst-case optimization problem (3) is almost the same as an empirical mean-variance problem. Specifically, if $\phi(z)$ satisfies (5) and is also twice continuously differentiable in the neighborhood of $z = 1$, then

$$\min_{\mathbb{Q}} \left\{ \mathbb{E}_{\mathbb{Q}}[f(x, Y)] + \frac{1}{\delta}\mathcal{H}_\phi(\mathbb{Q}\,|\,\mathbb{P}_n) \right\} = \mathbb{E}_{\mathbb{P}_n}[f(x, Y)] - \frac{\delta}{2\phi''(1)}\mathbb{V}_{\mathbb{P}_n}[f(x, Y)] + o(\delta), \tag{6}$$

where

$$\mathbb{V}_{\mathbb{P}_n}[f(x, Y)] := \frac{1}{n}\sum_{i=1}^n \left(f(x, Y_i) - \mathbb{E}_{\mathbb{P}_n}[f(x, Y_i)] \right)^2$$

is the variance of the reward under the empirical distribution. This shows that worst-case optimization (3) is actually a multiobjective problem that trades off between maximizing the expected reward $\mathbb{E}_{\mathbb{P}_n}[f(x, Y)]$ and minimizing its variance $\mathbb{V}_{\mathbb{P}_n}[f(x, Y)]$.

Although the mean-variance objective has been widely adopted, it is generally for reasons unrelated

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to model uncertainty, and it is natural to ask *why* it now appears in relation to a worst-case problem. It turns out that the variance of the reward measures worst-case sensitivity of the nominal model to misspecification. Intuitively, a decision x is sensitive to misspecification if small deviations from the nominal $\mathbb{P}_n$ can have a big impact on the expected reward $\mathbb{E}_{\mathbb{P}_n}[f(x, Y)]$. Sensitivity is large if the reward distribution has a large spread because perturbations that affect the tail can have a big impact on the mean. The worst-case optimizer (6) tries to optimize the expected reward while controlling the worst-case sensitivity to misspecification (spread).

This intuition can be formalized. Given a nominal model $\mathbb{P}_n$ and reward $f(x, Y)$, consider the family of distributions $\{\mathbb{Q}(\delta) : \delta \geq 0\}$, where

$$\mathbb{Q}(\delta) := \begin{cases} \arg\min_{\mathbb{Q}} \left\{ \sum_{i=1}^{n} q_i f(x, Y) + \frac{1}{\delta} \sum_{i=1}^{n} p_i^n \phi\left(\frac{q_i}{p_i^n}\right) \right\}, & \delta > 0, \\ \mathbb{P}_n, & \delta = 0. \end{cases} \tag{7}$$

Intuitively, $\{\mathbb{Q}(\delta) : \delta \geq 0\}$ is a family of worst-case deviations from the nominal, with the size of the deviation increasing in δ (corresponding to a smaller penalty on the ϕ-divergence in (7)). Although we have suppressed this from the notation, $\{\mathbb{Q}(\delta) : \delta \geq 0\}$ depends on the nominal $\mathbb{P}_n$ as well as the reward $f(x, Y)$. Given (7), we define the worst-case sensitivity of the expected reward $\mathbb{E}_{\mathbb{P}_n}[f(x, Y)]$ relative to nominal model $\mathbb{P}_n$ as

$$\begin{aligned} \mathcal{S}_{\mathbb{P}_n}(f(x, Y)) &:= -\frac{\mathrm{d}}{\mathrm{d}\delta} \mathbb{E}_{\mathbb{Q}(\delta)}[f(x, Y)]\Big|_{\delta=0} \\ &= -\lim_{\delta \downarrow 0} \frac{\mathbb{E}_{\mathbb{Q}(\delta)}[f(x, Y)] - \mathbb{E}_{\mathbb{P}_n}[f(x, Y)]}{\delta}. \end{aligned} \tag{8}$$

The following result shows that the worst-case sensitivity is equal to the variance. We require the following assumption.

Assumption 1. *The function $\phi : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ is a closed convex function such that $\phi(z) \geq \phi(1) = 0$ for $z \geq 0$, $\phi(z) = +\infty$ for $z < 0$, and it is twice continuously differentiable around $z = 1$ with $\phi''(1) > 0$.*

The proof of the following result can be found in Online Appendix Section EC.6.

Theorem 1. *Suppose $\phi(z)$ satisfies Assumption 1. Let $\{\mathbb{Q}(\delta) : \delta \geq 0\}$ be the family of worst-case measures defined by (7). Then the worst-case sensitivity of the expected reward $\mathbb{E}_{\mathbb{P}_n}[f(x, Y)]$ relative to the nominal $\mathbb{P}_n$ (8) satisfies*

$$\mathcal{S}_{\mathbb{P}_n}(f(x, Y)) = \frac{1}{\phi''(1)} \mathbb{V}_{\mathbb{P}_n}[f(x, Y)].$$

It can be also shown that the sensitivity of the expected reward under different models of misspecification (e.g., Wasserstein and CVaR) corresponds to different measures of spread and that the associated worst-case problems are mean-sensitivity problems. These results will be reported elsewhere. Worst-case sensitivity is also discussed in Lam (2016) for simulated estimates on an expectation when the family of alternative models is defined through a constraint on relative entropy. In this regard, Theorem 1 holds for all sufficiently smooth ϕ-divergence and not just for relative entropy. Variance penalties are proposed in Ban et al. (2018) as a way to control the sensitivity of solutions of an empirical mean CVaR portfolio choice problem to data variability, but the connection to worst-case optimization is not discussed.

2.1. Calibration

Robust optimization (3) defines a family of policies $\{x_n(\delta) : \delta \geq 0\}$ that balance between expected reward and sensitivity, with the weight on sensitivity increasing in δ. Our eventual goal is to identify values of the parameter δ such that the corresponding solution $x_n(\delta)$ performs well out of sample. Although parameters such as δ are commonly chosen by optimizing some estimate of the out-of-sample expected reward using bootstrap or cross-validation (e.g., the regularization parameter in Lasso or ridge regression), the characterization (6) shows that worst-case optimization is a multiobjective problem, which suggests that estimates of both the mean and the variance (i.e., sensitivity) should be used to select δ. We provide guidance for this approach by characterizing the impact of the robustness parameter on the mean and the variance of the out-of-sample reward.

2.2. The Constrained DRO Model

An alternative to the "penalty version" of the DRO model (3) defines a set of alternative models defined in terms of a hard constraint on the deviation measure:

$$\begin{cases} \max_x \min_{\mathbb{Q}} \mathbb{E}_{\mathbb{Q}}[f(x, Y)] \\ \text{subject to } \mathcal{H}_\phi(\mathbb{Q} \,|\, \mathbb{P}_n) \leq \epsilon. \end{cases} \tag{9}$$

As in the penalty version, this leads to a family of decisions $\{x_n(\epsilon) : \epsilon \geq 0\}$ parameterized by the constraint threshold ϵ in (9). With sufficient regularity, the penalty and constrained versions are related via convex duality in that the solution of (3) solves (9) for some choice of ϵ, and vice versa; see corollary 3 of Ben-Tal et al. (2013) for more details. Specifically, the solution of (9) is obtained by solving (3) with $\delta \equiv \delta_n(\epsilon)$:

$$x_n(\epsilon) = \arg\max_x \left\{ \min_{\mathbb{Q}} \mathbb{E}_{\mathbb{Q}}[f(x, Y)] + \frac{1}{\delta_n(\epsilon)} \mathcal{H}_\phi(\mathbb{Q} \,|\, \mathbb{P}_n) \right\} \tag{10}$$

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where

$$\delta_n(\epsilon) := \arg\max_{\delta\geq 0}\left\{\max_x \min_{\mathbb{Q}} \mathbb{E}_{\mathbb{Q}}\big[f(x,\,Y)\big] + \frac{1}{\delta}\mathcal{H}_\phi(\mathbb{Q}\,|\,\mathbb{P}_n) - \frac{\epsilon}{\delta}\right\}, \tag{11}$$

and both models generate the same family of solutions. For this reason, we focus on developing an approach to selecting δ in the penalty formulation of the model with the understanding that it can be applied to selecting ϵ in the constrained problem.

2.3. DRO Caveat

For the remainder of the paper, unless otherwise stated, all claims, findings, and insights provided are for DRO problems of the form (3). Conditions for the ϕ-divergence penalty and the reward function $f(x,\,Y)$ will be stated as needed.

3. Statistics of Robust Solutions: Asymptotic Normality

The results in Section 2 show that robust optimization is a multiobjective problem that balances between maximizing the expected reward and minimizing its variance. In this section, we characterize the statistical properties of the robust solution $x_n(\delta)$. These results will be used in the following sections to study the out-of-sample mean and variance (sensitivity) of the reward.

The results in this section use general results on the consistency (theorem 5.4 of Shapiro et al. 2014) and asymptotic normality (theorem 5.21 of van der Vaart 2000) of sample average optimizers, which are reproduced in Online Appendix Section EC.1. The following regularity assumptions on $f(x,\,Y)$ will generally be imposed. The results from Shapiro et al. (2014) and van der Vaart (2000) can be applied under more relaxed assumptions, though this does not lead to additional insights.

Let $f:\mathbb{R}^d\times\mathbb{R}^l\to\mathbb{R}$, let Y be an $\mathbb{R}^l$-valued random vector with distribution $\mathbb{P}$, and let

$$x^\star(0) := \arg\max_x\{\mathbb{E}_{\mathbb{P}}\big[f(x,\,Y)\big]\} \tag{12}$$

be the maximizer of the expected reward under the population distribution $\mathbb{P}$.

Assumption 2. *The function $f(x,\,Y)$ and random vector $Y\sim\mathbb{P}$ are such that*

- *$f(x,\,Y)$ is strictly concave and twice continuously differentiable in $x\in\mathbb{R}^d$ for $\mathbb{P}$-almost surely every $Y\in\mathbb{R}^l$;*
- *for each fixed $x\in\mathbb{R}^d$, the mappings $y\mapsto f(x,y)$, $\nabla_x f(x,y)$, $\nabla_x^2 f(x,y)$, $y\in\mathbb{R}^l$, are measurable, and all moments of the random variables $f(x,Y)$, $\nabla_x f(x,Y)$, and $\nabla_x^2 f(x,Y)$ exist; and*
- *there exists a solution $x^\star(0)$ of (12).*

It follows from Assumption 2 is that the Hessian matrix $\mathbb{E}_{\mathbb{P}}[\nabla_x^2 f(x,\,Y)]$ is negative definite for every x and that $x^\star(0)$ is unique. Observe, too, that Assumption 2 is not required for (6) to hold. Likewise, the relationship between worst-case optimization and mean-variance optimization still holds if x is constrained, though constraints will affect the efficiency of the mean-sensitivity trade-off for the reward that can be achieved.

The following result characterizes the asymptotic properties of the solution of the empirical optimization problem, and it follows immediately from results in Shapiro et al. (2014) and van der Vaart (2000), which are stated in Online Appendix Section EC.1.

Proposition 1. *Suppose that $f(x,\,Y)$ satisfies Assumption 2. Let $x^\star(0)$ be the solution of the optimization problem (12). Then the solution $x_n(0)$ of the empirical optimization problem (2) is consistent, $x_n(0)\xrightarrow{P}x^\star(0)$, and asymptotically normal:*

$$\sqrt{n}\big(x_n(0)-x^\star(0)\big)\xrightarrow{D}N(0,\,\xi(0)),\ \text{as } n\to\infty,$$

where the covariance matrix

$$\xi(0)=\mathbb{V}_{\mathbb{P}}\Big[\,\mathbb{E}_{\mathbb{P}}\big[\,\nabla_x^2 f\big(x^\star(0),\,Y\big)\big]^{-1}\nabla_x f\big(x^\star(0),\,Y\big)\Big].$$

In particular,

$$\sqrt{n}\big(x_n(0)-x^\star(0)\big)=\sqrt{\xi(0)}Z+o_P(1),$$

where $\sqrt{\xi(0)}$ is a $d\times d$ matrix such that $\sqrt{\xi(0)}\sqrt{\xi(0)}'=\xi(0)$, and Z is a d-dimensional standard normal random vector.

We now consider the asymptotic distribution of the solution $x_n(\delta)$ of the robust problem (3). The dual characterization of ϕ-divergence implies that $x_n(\delta)$ can be obtained by solving

$$(x_n(\delta),\,c_n(\delta))=\arg\min_{x,c}\left\{c+\frac{1}{\delta}\mathbb{E}_{\mathbb{P}_n}\Big[\phi^*\big(\delta\big(-f(x,\,Y)-c\big)\big)\Big]\right\}, \tag{13}$$

where

$$\phi^*(\zeta):=\sup_z\{z\zeta-\phi(z)\}$$

is the convex conjugate of $\phi(z)$. If $\psi:\mathbb{R}^d\times\mathbb{R}\to\mathbb{R}^{d+1}$ is given by

$$\psi(x,\,c):=\begin{bmatrix}(\phi^*)'\big(-\delta\big(f(x,\,Y)+c\big)\big)\nabla_x f(x,\,Y)\\ -\dfrac{\phi''(1)}{\delta}\Big((\phi^*)'\big(-\delta\big(f(x,\,Y)+c\big)\big)-1\Big)\end{bmatrix}, \tag{14}$$

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then the first-order conditions for $(x_n(\delta), c_n(\delta))$ are[1]

$$\begin{aligned}&\mathbb{E}_{\mathbb{P}_n}\big[\psi(x, c)\big]\\&\quad=\begin{bmatrix}\mathbb{E}_{\mathbb{P}_n}\big[(\phi^*)'\big(-\delta\big(f(x, Y)+c\big)\big)\nabla_x f(x, Y)\big]\\ \mathbb{E}_{\mathbb{P}_n}\Big[-\dfrac{\phi''(1)}{\delta}\Big((\phi^*)'\big(-\delta\big(f(x, Y)+c\big)\big)-1\Big)\Big]\end{bmatrix}\\&\quad=\begin{bmatrix}0\\0\end{bmatrix}.\end{aligned}\tag{15}$$

Similarly, let

$$\big(x^\star(\delta), c^\star(\delta)\big)=\arg\min_{x,c}\Big\{c+\frac{1}{\delta}\mathbb{E}_{\mathbb{P}}\big[\phi^*\big(\delta\big(-f(x, Y)-c\big)\big)\big]\Big\}\tag{16}$$

be the solution of the population version of the robust problem (13) with first-order conditions

$$\begin{aligned}&\mathbb{E}_{\mathbb{P}}\big[\psi(x, c)\big]\\&\quad=\begin{bmatrix}\mathbb{E}_{\mathbb{P}}\big[(\phi^*)'\big(-\delta\big(f(x, Y)+c\big)\big)\nabla_x f(x, Y)\big]\\ \mathbb{E}_{\mathbb{P}}\Big[-\dfrac{\phi''(1)}{\delta}\Big((\phi^*)'\big(-\delta\big(f(x, Y)+c\big)\big)-1\Big)\Big]\end{bmatrix}\\&\quad=\begin{bmatrix}0\\0\end{bmatrix}.\end{aligned}\tag{17}$$

Let $\xi(\delta)\in\mathbb{R}^{d\times d}$, $\eta(\delta)\in\mathbb{R}$, and $\kappa(\delta)\in\mathbb{R}^{d\times 1}$ be the entries of the matrix

$$V(\delta):=\begin{bmatrix}\xi(\delta) & \kappa(\delta)\\ \kappa(\delta)' & \eta(\delta)\end{bmatrix}=A^{-1}BA^{-1\prime},$$

where

$$\begin{aligned}A&:=\mathbb{E}_{\mathbb{P}}\big[-J_\psi\big(x^\star(\delta), c^\star(\delta)\big)\big]\in\mathbb{R}^{(d+1)\times(d+1)},\\B&:=\mathbb{E}_{\mathbb{P}}\big[\psi\big(x^\star(\delta), c^\star(\delta)\big)\psi\big(x^\star(\delta), c^\star(\delta)\big)'\big]\\&\quad\in\mathbb{R}^{(d+1)\times(d+1)},\end{aligned}$$

and J_ψ denotes the Jacobian matrix of ψ.

The first part of the following result shows that $(x_n(\delta), c_n(\delta))$ is consistent and jointly asymptotically normal, whereas the second part characterizes its limiting distribution in terms of the limiting distribution of the empirical distribution $x_n(0)$ when δ is small. Consistency and asymptotic normality can be established using the general results from Shapiro et al. (2014) and van der Vaart (2000) (Online Appendix Section EC.1). However, both $f(x, Y)$ and $\phi(x)$ need to be sufficiently regular, so in contrast to Proposition 1, Assumption 1 will be required in addition to Assumption 2.

Theorem 2. *Suppose $\phi(z)$ satisfies Assumption 1 and $f(x, Y)$ satisfies Assumption 2. Let $(x_n(\delta), c_n(\delta))$ be the solution of the robust empirical optimization problem (13) and $(x^\star(\delta), c^\star(\delta))$ solve the robust problem (16) under the population distribution $\mathbb{P}$. Then $(x_n(\delta), c_n(\delta))$ is consistent with $(x_n(\delta), c_n(\delta))\xrightarrow{P}(x^\star(\delta), c^\star(\delta))$ and jointly asymptotically normal with*

$$\begin{aligned}\sqrt{n}\big(x_n(\delta)-x^\star(\delta)\big)&\xrightarrow{D}\mathrm{N}(0, \xi(\delta)),\\\sqrt{n}\big(c_n(\delta)-c^\star(\delta)\big)&\xrightarrow{D}\mathrm{N}(0, \eta(\delta)),\\n\,\mathrm{Cov}_{\mathbb{P}}[x_n(\delta), c_n(\delta)]&\longrightarrow\kappa(\delta),\end{aligned}\tag{18}$$

as $n\to\infty$. In particular,

$$\sqrt{n}\big(x_n(\delta)-x^\star(\delta)\big)=\sqrt{\xi(\delta)}Z+o_P(1),$$

where $\sqrt{\xi(\delta)}$ is a $d\times d$ matrix such that $\sqrt{\xi(\delta)}\sqrt{\xi(\delta)}'=\xi(\delta)$, and Z is a d-dimensional standard normal random vector. Furthermore,

$$\begin{aligned}x^\star(\delta)&=x^\star(0)+\pi\delta+o(\delta),\\c^\star(\delta)&=-\mathbb{E}_{\mathbb{P}}\big[f\big(x^\star(0), Y\big)\big]+O(\delta),\end{aligned}\tag{19}$$

where

$$\begin{aligned}\pi:=&\frac{1}{\phi''(1)}\big(\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f\big(x^\star(0), Y\big)\big]\big)^{-1}\\&\times\mathrm{Cov}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(0), Y\big), f\big(x^\star(0), Y\big)\big],\end{aligned}\tag{20}$$

and

$$V(\delta)=\begin{bmatrix}\xi(\delta) & \kappa(\delta)\\ \kappa(\delta)' & \eta(\delta)\end{bmatrix}=\begin{bmatrix}\xi(0) & \kappa(0)\\ \kappa(0)' & \eta(0)\end{bmatrix}+O(\delta),$$

where

$$\begin{aligned}\xi(0)&=\mathbb{V}_{\mathbb{P}}\Big[\big(\mathbb{E}\big[\nabla_x^2 f\big(x^\star(0), Y\big)\big]\big)^{-1}\nabla_x f\big(x^\star(0), Y\big)\Big],\\\eta(0)&=\mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(0), Y\big)\big],\\\kappa(0)&=\big(\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f\big(x^\star(0), Y\big)\big]\big)^{-1}\\&\quad\times\mathrm{Cov}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(0), Y\big), f\big(x^\star(0), Y\big)\big].\end{aligned}$$

Proof of Theorem 2. Consistency and asymptotic normality (18) of $(x_n(\delta), c_n(\delta))$ follow from general results on the distributional properties of the solutions of sample average problems in Shapiro et al. (2014) and van der Vaart (2000); see Online Appendix Sections EC.1 and EC.2. To prove (19), we first show that $(x^\star(\delta), c^\star(\delta))$ is continuously differentiable in a neighborhood of $\delta=0$, following which we use the first-order conditions (15) for the optimization problem (13) to derive the Taylor series expansion (19) and the expression for the asymptotic bias (20). Details can be found in Online Appendix Section EC.3. □

Theorem 2 shows (asymptotically) that robust optimization adds a bias π to the empirical solution where the magnitude of the bias is determined by the robustness parameter δ. It can be shown that π optimizes a trade-off

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between the loss of expected reward and the reduction in variance:

$$\begin{aligned}\pi = \arg\max_{\pi}\Big\{&\frac{\delta^2}{2}\Big(\pi'\,\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f(x^\star(0),\,Y)\big]\pi \\ &-\frac{2}{\phi''(1)}\pi'\,\mathrm{Cov}_{\mathbb{P}}\big[f(x^\star(0),\,Y),\,\nabla_x f(x^\star(0),\,Y)\big]\Big) \\ &\equiv \mathbb{E}_{\mathbb{P}}\big[f(x^\star(0)+\delta\pi,\,Y)\big]-\mathbb{E}_{\mathbb{P}}\big[f(x^\star(0),\,Y)\big] \\ &-\frac{\delta}{2\phi''(1)}\big(\mathbb{V}_{\mathbb{P}}\big[f(x^\star(0)+\delta\pi,\,Y)\big]-\mathbb{V}_{\mathbb{P}}\big[f(x^\star(0),\,Y)\big]\big) \\ &+o(\delta^2)\Big\}.\end{aligned}$$

3.1. Variability of the Solution

It has been shown (Xu et al. 2010, Blanchet et al. 2017, Gao et al. 2017, Bertsimas and Copenhaver 2018) that various worst-case regression and classification problems are equivalent to regularized versions of these problems (i.e., regression/classification with ridge, Lasso, etc.). One implication is that robustness induces the bias-variance trade-off associated with regularization, leading to a reduction in the variance of the *solution* by shrinking it toward the origin. On the other hand, we showed in Theorem 2 that robustness adds a bias to the SAA solution in the direction π and changes its variance from $\xi(0)/n$ to $\xi(\delta)/n$. It is natural to ask whether this corresponds to shrinkage toward the origin and a reduction in the variance of the solution (perhaps when the reward function is concave).

The following example shows that this is not the case and that robustness can simultaneously bias the SAA solution away from the origin and increase its variance, even when the reward function is concave. More generally, (6) shows that DRO controls the variance (sensitivity) of the reward but is only equivalent to regularizing the solution in certain special cases.

Let

$$f(x,\,Y) = Yh(x) - x.$$

where the random variable Y is positive with distribution $\mathbb{P}$. We assume $h(x) = x^{-a}/(-a)$ $(a > 0)$, though we will only substitute this expression for $h(x)$ later in the example.

Given samples $Y_1, \cdots, Y_n \sim \mathbb{P}$ of Y, the solution $x_n(0)$ of the empirical problem (2) is the solution to the equation

$$\hat{Y}_n h'(x) - 1 = 0,$$

where the random variable $\hat{Y}_n = (Y_1 + \cdots + Y_n)/n$.

For concreteness, suppose that the deviation measure in the robust problem (4) is the modified χ^2 (see, e.g., Gotoh et al. (2018)). Writing $x_n(\delta) = x_n(0) + \pi_n\delta + o(\delta)$ for some π_n, the first-order conditions for (6) imply

$$x_n(\delta) = x_n(0) + \delta\left(\frac{\hat{\sigma}_n^Y}{\hat{Y}_n}\right)^2 \frac{h(x_n(0))}{h''(x_n(0))} + o(\delta),$$

where $\hat{\sigma}_n^Y$ is the sample standard deviation of Y and $\hat{\sigma}_n^Y/\hat{Y}_n$ is an estimate of its coefficient of variation. When $h(x)=x^{-a}/(-a)$, we have $h'(x) = x^{-a-1}$ and $h''(x) = -(a+1)x^{-a-2}$, so

$$x_n(\delta) = x_n(0) + \frac{\delta}{a(a+1)}x_n(0)^2\left(\frac{\hat{\sigma}_n^Y}{\hat{Y}_n}\right)^2 + o(\delta).$$

Observe that $x_n(0) = \hat{Y}^{\frac{1}{a+1}}$. In contrast to Lasso or ridge, which shrinks the sample average solution toward the origin, robustness introduces a positive bias that pushes it away.

The variance of the robust solution is given by

$$\begin{aligned}\mathbb{V}_{\mathbb{P}}[x_n(\delta)] = \mathbb{V}_{\mathbb{P}}[x_n(0)] &+ \frac{2\delta}{a(a+1)} \\ &\times \mathrm{Cov}_{\mathbb{P}}\left[x_n(0),\,x_n(0)^2\left(\frac{\hat{\sigma}_n^Y}{\hat{Y}_n}\right)^2\right] + o(\delta).\end{aligned}$$

Therefore, if the covariance of $x_n(0)$ and $x_n(0)^2(\frac{\hat{\sigma}_n^Y}{\hat{Y}_n})^2$ is positive (i.e., $\mathrm{Cov}_{\mathbb{P}}[x_n(0),\,x_n(0)^2(\frac{\hat{\sigma}_n^Y}{\hat{Y}_n})^2] > 0$), robustness increases the variance of the solution.

The upper plot in Figure 1 shows the distribution of $x_n(0)$ and $x_n(\delta)$ when Y is exponentially distributed with mean 1, $a = 0.05$, and there are $n = 10$ data points. The lower plot shows the objective function $\mathbb{E}_{\mathbb{P}}[f(x,\,Y)]$. When $\delta = 0.04$, the variance of the robust solution $x_n(\delta)$ is 0.8, which is larger than the variance of the solution of the empirical problem, which is 0.1.

4. Out-of-Sample Performance: Empirical Optimization

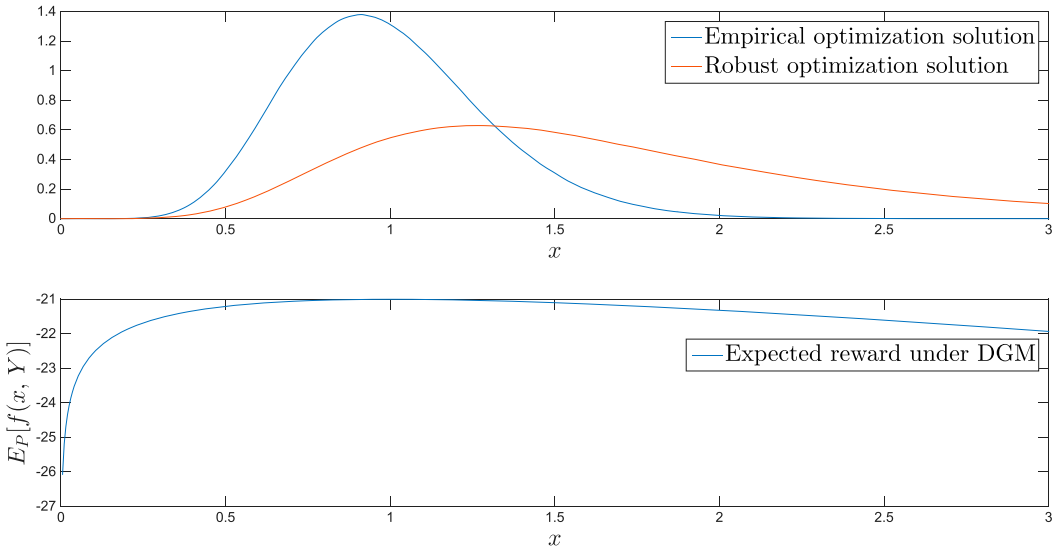
4.1. Preliminaries

Let $\mathcal{D} = \{Y_1, \cdots, Y_n\}$ be data generated i.i.d. under $\mathbb{P}$, let $x_n(0)$ be the solution of the empirical optimization problem (2) constructed using $\mathcal{D}$, and let Y_{n+1} be an additional sample generated under $\mathbb{P}$ independent of the original sample. Because $x_n(0)$ depends only the data $\mathcal{D}$ and is therefore independent of Y_{n+1} under the population distribution $\mathbb{P}$, it follows that $x_n(0) - x^\star(0)$ is also independent of Y_{n+1} under $\mathbb{P}$. Proposition 1 implies

$$x_n(0) = x^\star(0) + \sqrt{\frac{\xi(0)}{n}}(Z + o_P(1)). \tag{21}$$

In particular, $x_n(0)$ deviates from $x^\star(0)$ because of data variability, which is captured by the term involving Z and terms of $o_P(\frac{1}{\sqrt{n}})$, which for our purposes turn out to be negligible. We study the out-of-sample performance of $x_n(0)$ by evaluating the mean and variance of the

Figure 1. The Upper Plot Shows the Distribution of the Solutions of Empirical Optimization and Robust Optimization When There Are $n = 10$ Data Points, and the Lower Plot Shows the Objective Function Under the Data Generating Mechanism (DGM)



Notes. The optimal solution is $x^\star = 1$. In this example, robustness adds a bias of 0.75 to the empirical solution and pushes it away from the origin. Robustness also increases the variance of the solution from 0.1 to 0.8.

reward $f(x_n(0), Y_{n+1})$ over random samples of the data $\mathcal{D}$ and Y_{n+1} generated under $\mathbb{P}$.

Our subsequent results characterize the impact of robustness through a Taylor series expansion of the mean and variance of the out-of-sample reward when the robustness parameter is small. The following result is a general statement of this expansion, which will streamline some of these derivations. The smoothness and integrability conditions are the same as those stated in Assumption 2 and guarantee that Taylor series expansions are defined and expected values and variances exist. The proof of this result can be found in Online Appendix Section EC.4.

Proposition 2. *Suppose that $f(x, Y)$ is twice continuously differentiable in $x \in \mathbb{R}^d$ for $\mathbb{P}$-almost surely every $Y \in \mathbb{R}^l$; that for every fixed $x \in \mathbb{R}^d$, the mappings $y \mapsto f(x, y)$, $\nabla_x f(x, y), \nabla_x^2 f(x, y)$, $y \in \mathbb{R}^l$, are measurable; and that all moments of the random variables $f(x, Y), \nabla_x f(x, Y), \nabla_x^2 f(x, Y)$ exist where Y has distribution $\mathbb{P}$. Let $Y_1, \ldots, Y_n, Y_{n+1}$ be independently generated from the distribution $\mathbb{P}$. Let $x \in \mathbb{R}^d$ be constant, and let $\Delta \in \mathbb{R}^d$ a random vector independent of Y_{n+1} under $\mathbb{P}$. Then for any constant δ,*

$$\begin{aligned}\mathbb{E}_{\mathbb{P}}\big[f(x+\delta\Delta, Y_{n+1})\big] &= \mathbb{E}_{\mathbb{P}}\big[f(x, Y_{n+1})\big] \\ &+ \delta\, \mathbb{E}_{\mathbb{P}}[\Delta]'\, \mathbb{E}_{\mathbb{P}}\big[\nabla_x f(x, Y_{n+1})\big] \\ &+ \frac{\delta^2}{2}\operatorname{tr}\big(\mathbb{E}_{\mathbb{P}}[\Delta\Delta']\, \mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f(x, Y_{n+1})\big]\big) + o(\delta^2),\end{aligned} \tag{22}$$

$$\begin{aligned}\mathbb{V}_{\mathbb{P}}\big[f(x+\delta\Delta, Y_{n+1})\big] &= \mathbb{V}_{\mathbb{P}}\big[f(x, Y_{n+1})\big] \\ &+ \delta \mathbb{E}_{\mathbb{P}}[\Delta]'\, \nabla_x \mathbb{V}_{\mathbb{P}}\big[f(x, Y_{n+1})\big] \\ &+ \frac{\delta^2}{2}\operatorname{tr}\big(\mathbb{E}_{\mathbb{P}}[\Delta\Delta']\, \nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f(x, Y_{n+1})\big] + 2\mathbb{V}_{\mathbb{P}}[\Delta] \\ &\times \mathbb{E}_{\mathbb{P}}\big[\nabla_x f(x, Y_{n+1})\big]\, \mathbb{E}_{\mathbb{P}}\big[\nabla_x f(x, Y_{n+1})\big]'\big) + o(\delta^2),\end{aligned} \tag{23}$$

where the first and second derivatives of the variance of the reward with respect to decision x satisfy

$$\nabla_x \mathbb{V}_{\mathbb{P}}\big[f(x, Y_{n+1})\big] = 2\operatorname{Cov}_{\mathbb{P}}\big[f(x, Y_{n+1}), \nabla_x f(x, Y_{n+1})\big], \tag{24}$$

$$\begin{aligned}\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f(x, Y_{n+1})\big] &= 2\,\mathbb{V}_{\mathbb{P}}\big[\nabla_x f(x, Y_{n+1})\big] \\ &+ 2\operatorname{Cov}_{\mathbb{P}}\big[f(x, Y_{n+1}), \nabla_x^2 f(x, Y_{n+1})\big].\end{aligned} \tag{25}$$

4.2. Empirical Optimization

We now derive an expansion of the out-of-sample expected reward and variance under the empirical optimal solution, which will serve as a baseline case when studying the out-of-sample performance of robust optimization.

To ease notation, we denote the mean and variance of the out-of-sample reward under $x_n(\delta)$ and the population solution $x^\star(\delta)$ as

$$\begin{aligned}&\mu_n(\delta) := \mathbb{E}_{\mathbb{P}}\big[f(x_n(\delta), Y_{n+1})\big],\ v_n(\delta) := \mathbb{V}_{\mathbb{P}}\big[f(x_n(\delta), Y_{n+1})\big], \\ &\mu^\star(\delta) := \mathbb{E}_{\mathbb{P}}\big[f(x^\star(\delta), Y_{n+1})\big],\ v^\star(\delta) := \mathbb{V}_{\mathbb{P}}\big[f(x^\star(\delta), Y_{n+1})\big],\end{aligned} \tag{26}$$

($\delta = 0$ corresponds to empirical optimization). The curvature of the expected reward and the second derivative of the variance of the reward at $x^\star(\delta)$ also appear. We denote these as

$$\begin{aligned}H(\delta) &:= \mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f(x^\star(\delta), Y_{n+1})\big], \\ G(\delta) &:= \nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f(x^\star(\delta), Y_{n+1})\big],\end{aligned} \tag{27}$$

where $\nabla_x^2 \mathbb{V}_{\mathbb{P}}[f(x^\star(\delta), Y_{n+1})]$ is defined in (25).

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The mean and variance of the out-of-sample reward depend on the variability of the empirical solution $x_n(0)$ and the properties of the reward function. The following result relates the mean and variance of the out-of-sample reward for the empirical optimizer to those of the population optimizer $x^\star(0)$.

Proposition 3. *Suppose that $f(x, Y)$ satisfies Assumption 2. Then the mean and variance of the out-of-sample reward under the empirical solution $x_n(0)$ satisfy*

$$\mu_n(0) = \mu^\star(0) + \frac{1}{2n}\operatorname{tr}(\xi(0)H(0)) + o\left(\frac{1}{n}\right), \tag{28}$$

$$v_n(0) = v^\star(0) + \frac{1}{2n}\operatorname{tr}(\xi(0)\,G(0)) + o\left(\frac{1}{n}\right), \tag{29}$$

where $\mu^\star(0)$ and $v^\star(0)$ are the out-of-sample mean and variance of the population optimizer $x^\star(0)$, respectively.

Proof of Proposition 3. It is clear from (26) that the mean and variance of the out-of-sample reward depend on the distribution properties of the solution $x_n(0)$, the reward function, and the distribution of Y_{n+1}. For the solution, Proposition 1 shows that (21) holds under Assumption 2, and hence

$$\begin{aligned} v_n(0) &= \mathbb{V}_{\mathbb{P}}\left[f(x_n(0), Y_{n+1})\right] \\ &= \mathbb{V}_{\mathbb{P}}\left[f\left(x^\star(0) + \sqrt{\frac{\xi(0)}{n}}(Z + o_P(1)), Y_{n+1}\right)\right]. \end{aligned}$$

This can be evaluated using Proposition 2 (with $\Delta = \sqrt{\xi(0)}Z$ and $\delta = \frac{1}{\sqrt{n}}$) (see Online Appendix Section EC.5 for more details), from which the expression (29) follows. The expression (28) for the expected out-of-sample reward under the empirical optimal can be derived in the same way. □

Proposition 3 quantifies the difference between the out-of-sample means and out-of-sample variances for the solution $x^\star(0)$ of the population problem and the empirical solution $x_n(0)$. In particular, (28) shows that the difference between the expected rewards is increasing in the variance $\frac{\xi(0)}{n}$ of the solution $x_n(0)$ and the curvature $H(0)$ of the expected reward around $x^\star(0)$:

$$\mu^\star(0) - \mu_n(0) = -\frac{1}{2n}\operatorname{tr}(\xi(0)H(0)) + o\left(\frac{1}{n}\right) > 0. \tag{30}$$

This performance gap can be attributed to Jensen's inequality. Specifically, if

$$F(x) = \mathbb{E}_{\mathbb{P}}\{f(x, Y_{n+1}) \,|\, x\}$$

denotes the out-of-sample expected reward, then $F(x)$ is strictly concave in x (Assumption 2), and

$$\begin{aligned} F(\mathbb{E}_{\mathbb{P}}[x_n(0)]) - \mathbb{E}_{\mathbb{P}}[F(x_n(0))] &= \mathbb{E}_{\mathbb{P}}\left[f(\mathbb{E}_{\mathbb{P}}[x_n(0)], Y_{n+1})\right] \\ &\quad - \mathbb{E}_{\mathbb{P}}\left[\mathbb{E}_{\mathbb{P}}\{f(x_n(0), Y_{n+1}) \,|\, x_n(0)\}\right] > 0 \end{aligned}$$

by Jensen's inequality. Because $\mathbb{E}_{\mathbb{P}}[x_n(0)] = x^\star(0) + o(1/\sqrt{n})$,

$$\begin{aligned} &F(\mathbb{E}_{\mathbb{P}}[x_n(0)]) - \mathbb{E}_{\mathbb{P}}[F(x_n(0))] \\ &= \mathbb{E}_{\mathbb{P}}\left[f(\mathbb{E}_{\mathbb{P}}[x_n(0)], Y_{n+1})\right] \\ &\qquad - \mathbb{E}_{\mathbb{P}}\left[\mathbb{E}_{\mathbb{P}}\{f(x_n(0), Y_{n+1}) \,|\, x_n(0)\}\right] + o\left(\frac{1}{n}\right) \\ &= -\frac{1}{2n}\operatorname{tr}\left(\xi(0)\,\mathbb{E}_{\mathbb{P}}\left[\nabla_x^2 f(x^\star(0), Y_{n+1})\right]\right) + o\left(\frac{1}{n}\right), \end{aligned}$$

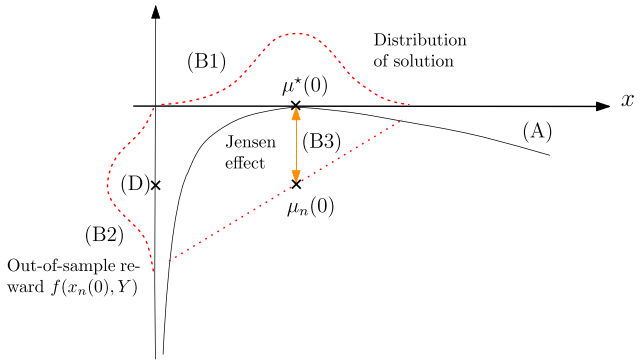
which is precisely the performance gap in (28) (or equivalently, (30)).

An illustration of Proposition 3 is shown in Figure 2, where (A) is the expected reward $F(x)$ under the population distribution on which × is the optimal value $\mu^\star(0)$; (B1) is the distribution of $x_n(0)$, and (B2) is the distribution of the out-of-sample reward $f(x_n(0), Y)$. The expected out-of-sample reward $\mu_n(0)$ is marked × at (D) and at the point labelled $\mu_n(0)$. We can see that when $F(x)$ is strictly concave, variability in $x_n(0)$ around its asymptotic mean $x^\star(0)$ leads to a gap between $\mu^\star(0)$ and $\mu_n(0)$ from Jensen's inequality that is shown at (B3). This gap is quantified in Proposition 3.

5. Out-of-Sample Performance: Robust Optimization

We now study the mean and variance of the out-of-sample reward generated by solutions of robust optimization problems $x_n(\delta)$ and characterize the impact of the robustness parameter δ. Our analysis consists of two parts. To begin, we derive expressions analogous to (28) and (29) in which the out-of-sample mean (variance) of the reward $f(x_n(\delta), Y_{n+1})$ is decomposed into the mean (variance) of the reward associated with the solution $x^\star(\delta)$ of the population problem and an additional component that is determined by the variability of the solution around the asymptotic mean and

Figure 2. Illustration of Proposition 3



Notes. The plot (A) is the expected reward $F(x)$ under the population distribution. (B1) is the distribution of $x_n(0)$, (B2) is the distribution of the out-of-sample reward $f(x_n(0), Y)$ under $x_n(0)$, and (B3) shows the gap $\mu^\star(0) - \mu_n(0)$ from Jensen's inequality given by (30). The expected reward $\mu_n(0)$ is shown at (D), and $\mu^\star(0)$ is indicated on (A).

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the curvature of the reward function; see (32) and (38). Next, we evaluate the impact of the robustness parameter on the mean and variance of the out-of-sample reward.

5.1. Expected Reward

We analyze the impact of robustness on the mean of the out-of-sample reward by expanding $\mu_n(\delta) = \mathbb{E}_{\mathbb{P}}[f(x_n(\delta), Y_{n+1})]$ around $\delta = 0$.

To begin, we evaluate the impact of solution variability and the curvature of the expected reward on the out-of-sample expected reward under the robust solution. Suppose that $\phi(z)$ satisfies Assumption 1 and $f(x, Y)$ satisfies Assumption 2. Recall from Theorem 2 that the robust solution is asymptotically normal:

$$x_n(\delta) = x^\star(\delta) + \sqrt{\frac{\xi(\delta)}{n}}(Z + o_P(1)). \tag{31}$$

It follows that the out-of-sample expected reward is

$$\begin{aligned}
&\overbrace{\mathbb{E}_{\mathbb{P}}\big[f(x_n(\delta), Y_{n+1})\big]}^{\mu_n(\delta)} \\
&= \mathbb{E}_{\mathbb{P}}\left[f\left(x^\star(\delta) + \frac{\sqrt{\xi(\delta)}}{\sqrt{n}}(Z + o_P(1)), Y_{n+1}\right)\right] \\
&= \underbrace{\mathbb{E}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big]}_{\mu^\star(\delta)} + \underbrace{\frac{1}{2n}\mathrm{tr}\big(\xi(\delta)\,\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f\big(x^\star(\delta), Y_{n+1}\big)\big]\big)}_{\frac{1}{2n}\mathrm{tr}(\xi(\delta)\,H(\delta))\ \equiv \text{Jensen effect}} \\
&\quad + o\left(\frac{1}{n}\right).
\end{aligned} \tag{32}$$

Noting (21), robustness changes the mean of the solution from $x^\star(0)$ to $x^\star(\delta)$ and its asymptotic variance from $\xi(0)/n$ to $\xi(\delta)/n$. The expression (32) is analogous to (28) but reflects the new mean and variance of the solution. Specifically, the first term is the expected reward under asymptotic mean $x^\star(\delta)$, and the second term is the Jensen effect—namely, the reduction in the expected reward $\mathbb{E}_{\mathbb{P}}[f(x^\star(\delta), Y_{n+1})] - \mathbb{E}_{\mathbb{P}}[f(x_n(\delta), Y_{n+1})]$ as a result of fluctuations of the robust solution $x_n(\delta)$ around the asymptotic mean $x^\star(\delta)$ and the curvature of the expected reward.

To understand the impact of robustness on the out-of-sample expected reward, we expand both terms in (32) around $\delta = 0$ (i.e., the SAA solution). By (19) and Proposition 2 (with $\Delta = \pi$), we have

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big] &= \mathbb{E}_{\mathbb{P}}\big[f\big(x^\star(0), Y_{n+1}\big)\big] \\
&\quad + \frac{\delta^2}{2}\pi'\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f\big(x^\star(0), Y_{n+1}\big)\big]\pi + o\big(\delta^2\big) \\
&= \mu^\star(0) + \frac{\delta^2}{2}\pi' H(0)\pi + o\big(\delta^2\big).
\end{aligned} \tag{33}$$

Robustness changes the asymptotic mean of the solution $x_n(\delta)$ from $x^\star(0)$ to $x^\star(\delta) = x^\star(0) + \delta\pi + o(\delta)$, which changes the expected reward. However, (33) shows that this change is of order δ^2, which is small when δ is small.

In the case of the Jensen effect, we have

$$\begin{aligned}
&\frac{1}{2n}\mathrm{tr}(\xi(\delta)H(\delta)) \\
&= \underbrace{\frac{1}{2n}\mathrm{tr}(\xi(0)H(0))}_{\text{Jensen effect for } x_n(0)} + \underbrace{\frac{1}{2n}\mathrm{tr}((\xi(\delta) - \xi(0))H(0))}_{\text{Change in variance of solution}} \\
&\quad + \underbrace{\frac{1}{2n}\mathrm{tr}(\xi(0)(H(\delta) - H(0)))}_{\text{Change in curvature}} \\
&\quad + \underbrace{\frac{1}{2n}\mathrm{tr}((\xi(\delta) - \xi(0))(H(\delta) - H(0)))}_{\text{Negligible}}.
\end{aligned} \tag{34}$$

The first term is the Jensen effect associated with the empirical solution (28), whereas the remaining terms are adjustments to the Jensen effect for $x_n(0)$ as a result of the change in the variance of the solution from $\xi(0)/n$ to $\xi(\delta)/n$ and the change in the curvature as a result of the shift in the mean of the solution from $x^\star(0)$ to $x^\star(\delta)$. If $f(x, Y)$ is such that for any matrix $Q \in \mathbb{R}^{d\times d}$, the function $L : \mathbb{R}^d \to \mathbb{R}$, where

$$L(x) := \mathrm{tr}\big(Q\,\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f(x, Y_{n+1})\big]\big) \tag{35}$$

is continuously differentiable, it follows from the definition (27) of $H(\delta)$ and the expansion (19) of $x^\star(\delta)$ that

$$\frac{1}{2n}\mathrm{tr}(\xi(\delta)H(\delta)) = \frac{1}{2n}\mathrm{tr}(\xi(0)H(0)) + \frac{\rho\delta}{2n} + o(\delta), \tag{36}$$

where

$$\begin{aligned}
\rho &:= \mathrm{tr}(\xi'(0)H(0)) \\
&\quad + \pi'\nabla_x\big[\mathrm{tr}\big(\xi(0)\,\mathbb{E}_{\mathbb{P}}\big[\nabla_x^2 f\big(x^\star(0), Y_{n+1}\big)\big]\big)\big],
\end{aligned} \tag{37}$$

where $\xi'(0)$ is the derivative of $\xi(\delta)$ with respect to δ evaluated at $\delta = 0$ (Theorem 2). Compared with (33), where the impact of robustness is $O(\delta^2)$, the impact of robustness on the Jensen effect (36) is $O(\delta)$, coming from the adjustment $\frac{\rho\delta}{2n}$. This term is scaled by a factor $1/n$, so although it is linear in δ, the net effect is small when δ is small. The expansion of the out-of-sample expected reward is obtained by combining (32), (33), and (36).

5.2. Variance of the Reward

We analyze the impact of robustness on the variance of the out-of-sample reward by expanding $v_n(\delta) = \mathbb{V}_{\mathbb{P}}[f(x_n(\delta), Y_{n+1})]$ around $\delta = 0$.

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Under Assumptions 1 and 2, it is shown in Theorem 2 that robustness changes the mean of the solution from $x^\star(0)$ to $x^\star(\delta)$ and its asymptotic variance from $\xi(0)/n$ to $\xi(\delta)/n$ (see also (21) and (31)). To see the impact on the variance of the out-of-sample reward, observe from (23) that

$$\begin{aligned}
&\mathbb{V}_{\mathbb{P}}\big[f(x_n(\delta), Y_{n+1})\big] \\
&= \mathbb{V}_{\mathbb{P}}\left[f\left(x^\star(\delta) + \frac{\sqrt{\xi(\delta)}}{\sqrt{n}}(Z + o_P(1)), Y_{n+1}\right)\right] \\
&= \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big] + \frac{1}{2n}\mathrm{tr}\Big(\xi(\delta)\Big\{\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big] \\
&\quad + 2\mathbb{E}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(\delta), Y_{n+1}\big)\big]\mathbb{E}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(\delta), Y_{n+1}\big)\big]'\Big\}\Big) + o\left(\frac{1}{n}\right),
\end{aligned} \tag{38}$$

where $\nabla_x^2 \mathbb{V}_{\mathbb{P}}[f(x^\star(\delta), Y_{n+1})]$ is given in (25). The first term in (38) is the variance of the reward $v^\star(\delta)$ at the optimal solution $x^\star(\delta)$ of the population robust model. The second term is the impact of the variability of the robust solution $x_n(\delta)$ on the variance of the reward. The expansion (38) is analogous to the expansion (29) of the variance for the empirical problem, the difference being that it is for the robust solution $x_n(\delta)$ instead of $x_n(0)$.

To see the impact of robustness on the variance of the reward, we expand (38) around $\delta = 0$. Noting (19), it follows from Proposition 2 that the first term

$$\begin{aligned}
\mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big] &= \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(0), Y_{n+1}\big)\big] \\
&\quad + \delta\pi'\big(\nabla_x \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(0), Y_{n+1}\big)\big]\big),
\end{aligned} \tag{39}$$

where $\nabla_x \mathbb{V}_{\mathbb{P}}[f(x^\star(0), Y_{n+1})]$ is defined in (24).

As for the second term in (38), observe that

$$\begin{aligned}
&\frac{1}{2n}\mathrm{tr}\big(\xi(\delta)\,\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big]\big) \\
&= \frac{1}{2n}\mathrm{tr}(\xi(0)G(0)) + \frac{1}{2n}\mathrm{tr}((\xi(\delta) - \xi(0))G(0)) \\
&\quad + \frac{1}{2n}\mathrm{tr}(\xi(0)(G(\delta) - G(0))) \\
&\quad + \frac{1}{2n}\mathrm{tr}((\xi(\delta) - \xi(0))(G(\delta) - G(0))).
\end{aligned}$$

If $f(x, Y)$ is such that for any matrix $Q \in \mathbb{R}^{d\times d}$, the function $M : \mathbb{R}^d \to \mathbb{R}$, where

$$M(x) := \mathrm{tr}\big(Q\,\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f(x, Y_{n+1})\big]\big) \tag{40}$$

is continuously differentiable, the expansion (19) of $x^\star(\delta)$ and the definition of $G(\delta)$ in (27) implies

$$\begin{aligned}
&\frac{1}{2n}\mathrm{tr}\big(\xi(\delta)\,\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big]\big) \\
&\quad = \frac{1}{2n}\mathrm{tr}(\xi(0)G(0)) + \frac{\delta}{2n}\mathrm{tr}(\xi'(0)G(0)) \\
&\quad + \frac{\delta}{2n}\pi'\nabla_x\Big[\mathrm{tr}\big(\xi(0)\,\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(0), Y_{n+1}\big)\big]\big)\Big] + \frac{1}{2n}O(\delta^2).
\end{aligned}$$

Because

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(\delta), Y_{n+1}\big)\big]\mathbb{E}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(\delta), Y_{n+1}\big)\big]' \\
&\qquad \sim O(\delta^2),
\end{aligned}$$

it follows that

$$\begin{aligned}
&\frac{1}{2n}\mathrm{tr}\Big(\xi(\delta)\Big\{\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(\delta), Y_{n+1}\big)\big] \\
&\quad + 2\mathbb{E}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(\delta), Y_{n+1}\big)\big]\,\mathbb{E}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(\delta), Y_{n+1}\big)\big]'\Big\}\Big) \\
&\quad = \frac{1}{2n}\mathrm{tr}(\xi(0)G(0)) + \frac{\theta\delta}{2n} + \frac{1}{n}O(\delta^2),
\end{aligned} \tag{41}$$

where

$$\begin{aligned}
\theta &:= \mathrm{tr}(\xi'(0)G(0)) \\
&\quad + \pi'\nabla_x\big[\mathrm{tr}\big(\xi(0)\,\nabla_x^2 \mathbb{V}_{\mathbb{P}}\big[f\big(x^\star(0), Y_{n+1}\big)\big]\big)\big].
\end{aligned} \tag{42}$$

The first term in (41) is from the empirical optimizer (29), whereas the term $\frac{\theta\delta}{2n}$ reflects the impact of robustness on the out-of-sample variance. An expansion of the out-of-sample variance $v_n(\delta) = \mathbb{V}_{\mathbb{P}}[f(x_n(\delta), Y_{n+1})]$ around $\delta = 0$ is obtained by combining (39) and (41).

5.3. Main Result

Let

$$\big\{\big(\mu_n(\delta), v_n(\delta)\big) : \delta \geq 0\big\} \tag{43}$$

be the robust mean-variance frontier (26). The following result characterizes the frontier in the asymptotic regime when n is large and δ is small. In addition to Assumption 2, the following assumption is required.

Assumption 3. *The function $f(x, Y)$ is such that for every matrix $Q \in \mathbb{R}^{d\times d}$, the functions $L : \mathbb{R}^d \to \mathbb{R}$ and $M : \mathbb{R}^d \to \mathbb{R}$ defined in (35) and (40), respectively, are continuously differentiable in $x \in \mathbb{R}^d$.*

Proposition 4. *Suppose $\phi(z)$ satisfies Assumption 1 and $f(x, Y)$ satisfies Assumptions 2 and 3. Then the expected value of the out-of-sample reward under the robust solution $x_n(\delta)$ is*

$$\begin{aligned}
\mu_n(\delta) &= \mu^\star(0) + \frac{1}{2n}\mathrm{tr}(\xi(0)H(0)) \\
&\quad + \frac{\delta^2}{2\big[\phi''(1)\big]^2}\beta' H(0)^{-1}\beta + \frac{\rho\delta}{2n} \\
&\quad + \frac{1}{n}O(\delta^2) + o\left(\frac{1}{n}\right) + o(\delta^2),
\end{aligned} \tag{44}$$

where $H(0)$ is the curvature of the out-of-sample expected reward at $x^\star(0)$ as defined in (27), and

$$\beta := \mathrm{Cov}_{\mathbb{P}}\big[\nabla_x f\big(x^\star(0), Y_{n+1}\big), f\big(x^\star(0), Y_{n+1}\big)\big]. \tag{45}$$

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The variance of the out-of-sample reward is

$$v_n(\delta) = v^\star(0) + \frac{1}{2n}\operatorname{tr}(\xi(0)G(0)) + \frac{2\delta}{\phi''(1)}\beta' H(0)^{-1}\beta + \frac{\theta\delta}{2n} + \frac{1}{n}O(\delta^2) + o\left(\frac{1}{n}\right) + O(\delta^2). \tag{46}$$

Proof of Proposition 4. It follows from (32), (33), and (36) and the expression (20) for π that the out-of-sample expected reward is given by (44). The out-of-sample variance is given by (38), (39), and (41). □

For some perspective on Proposition 4, note that the mean and variance of the reward under the solution $x^\star(0)$ of the robust population problem are given by

$$\mu^\star(\delta) = \mu^\star(0) + \frac{\delta^2}{2[\phi''(1)]^2}\beta' H(0)^{-1}\beta + o(\delta^2),$$
$$v^\star(\delta) = v^\star(0) + \frac{2\delta}{\phi''(1)}\beta' H(0)^{-1}\beta + O(\delta^2), \tag{47}$$

(we also get these equations when we let $n \to \infty$ in (44) and (46)), from which it follows that

$$\frac{d\mu^\star}{d\delta}(0) = 0,$$
$$\frac{dv^\star}{d\delta}(0) = \frac{2}{\phi''(1)}\beta' H(0)^{-1}\beta'.$$

Because $f(x, Y)$ is strictly concave in x, the curvature $H(0)$ is negative definite, so variance decreases linearly around $\delta = 0$. In fact, (47) shows that both the mean and variance are decreasing in δ, but variance reduction is an order of magnitude larger than the reduction in the mean when δ is small ($O(\delta)$ versus $O(\delta^2)$). As δ increases, however, the quadratic term becomes significant, and the mean decreases at an increasing rate. This suggests that the choice of δ has a significant impact on the out-of-sample reward and that calibration of δ should therefore account for both the mean and the variance.

When n is finite, the robust mean-variance frontier $\{(\mu_n(\delta), v_n(\delta)) : \delta \geq 0\}$ deviates from the population frontier (47) because the solution $x_n(\delta)$ is a random variable. Indeed, (44) and (46) can be written as

$$\mu_n(\delta) = \mu^\star(\delta) + \overbrace{\frac{1}{2n}\operatorname{tr}(\xi(0)H(0)) + \frac{\rho\delta}{2n}}^{\text{Jensen effect}} + \overbrace{\frac{1}{n}O(\delta^2) + o\left(\frac{1}{n}\right) + o(\delta^2)}^{\text{higher-order terms}},$$
$$v_n(\delta) = v^\star(\delta) + \frac{1}{2n}\operatorname{tr}(\xi(0)G(0)) + \frac{\theta\delta}{2n} + \frac{1}{n}O(\delta^2) + o\left(\frac{1}{n}\right) + O(\delta^2),$$

showing that difference in out-of-sample expected rewards $\mu_n(\delta) - \mu^\star(\delta)$ comes from the variability in $x_n(\delta)$ via the Jensen effect, with a similar interpretation for the variance. The first term in the Jensen effect is from the variability of $x_n(0)$ around $x^\star(0)$ (see (28)), wheras the second term is the modification to the Jensen effect as a result of the bias that is added to the solution by robustness, which was discussed in (36) and (37).

Another interpretation of Proposition 4 can be obtained by writing the robust mean-variance frontier $(\mu_n(\delta), v_n(\delta))$ in terms the mean and variance $(\mu_n(0), v_n(0))$ of the out-of-sample reward for the empirical optimal $x_n(0)$, which was derived in Proposition 3.

Theorem 3. *Suppose $\phi(z)$ satisfies Assumption 1 and $f(x, Y)$ satisfies Assumptions 2 and 3. Let the constants ρ and θ be defined by (37) and (42) and $H(0)$ and β by (27) and (45). Then*

$$\mu_n(\delta) = \mu_n(0) + \frac{\rho\delta}{2n} + \frac{\delta^2}{2[\phi''(1)]^2}\beta' H(0)^{-1}\beta + \frac{1}{n}O(\delta^2) + o(\delta^2), \tag{48}$$
$$v_n(\delta) = v_n(0) + \delta\left[\frac{2}{\phi''(1)}\beta' H(0)^{-1}\beta + \frac{\theta}{2n}\right] + \frac{1}{n}O(\delta^2) + O(\delta^2). \tag{49}$$

In this case,

$$\frac{d\mu_n}{d\delta}(0) = \frac{\rho}{2n},$$
$$\frac{dv_n}{d\delta}(0) = \frac{2}{\phi''(1)}\beta' H(0)^{-1}\beta + \frac{\theta}{2n},$$

so, in contrast the population frontier, derivatives are nonzero at $\delta = 0$. In the case of the variance, $\beta' H(0)^{-1}\beta < 0$. Because the term $\frac{\theta}{2n}$ is scaled by a factor $1/n$, for n sufficiently large, the derivative is negative, and variance reduces linearly about $\delta = 0$. The rate of change of the mean is also linear, but for n sufficiently large, $\frac{\rho}{2n}$ is small, and variance reduction will dominate. As δ increases, the quadratic term in (48) eventually dominates, at which point the mean decreases at an increasing rate because the coefficient is negative. Interestingly, the marginal change in the mean is equal to the adjustment to the Jensen effect from robustness (see (34), (36), and (37)).

In summary, (48) and (49) show that when δ is small and n large, the reduction in the variance of the out-of-sample reward under $x_n(\delta)$ is significantly larger than the impact on the mean. These results apply when x is unconstrained. If x is constrained, the mean-sensitivity relationship (6) still holds, so worst-case optimization is still making a trade-off between expected reward and sensitivity, but the constraints will affect the efficiency that can be achieved.

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5.4. Expected Reward of DRO Can Outperform SAA

The term $\frac{\rho}{2n}\delta$ in (48) is related to the Jensen effect. In particular, robustness changes the mean of the empirical solution from $x_n(0)$ to $x_n(\delta)$ and its variance from $\xi(0)$ to $\xi(\delta)$, and it was shown in (36) and (37) that this changes the Jensen effect from

$$\frac{1}{2n}\mathrm{tr}(\xi(0)H(0))$$

for the empirical solution $x_n(0)$ to

$$\frac{1}{2n}\mathrm{tr}(\xi(0)H(0)) + \frac{\rho}{2n}\delta.$$

Note that ρ can be positive or negative. It is positive if the bias $\delta\pi$ that is introduced by robustness reduces the loss from the Jensen effect. When this happens, $\mu_n(\delta)$ is larger than $\mu_n(0)$ when δ is sufficiently small for the linear term in (48) to dominate the quadratic. That is, although DRO is a worst-case problem, the robust solution can have a larger out-of-sample expected reward than the SAA solution.

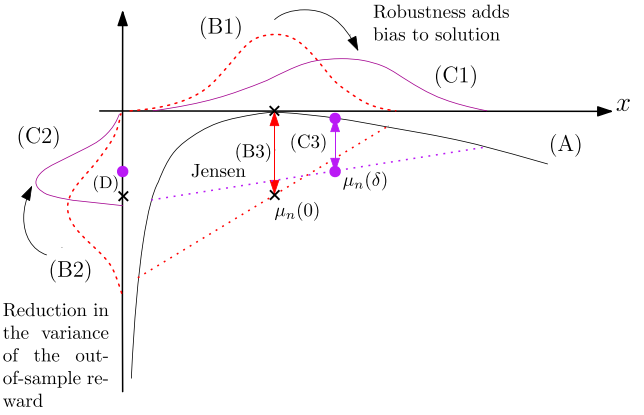
Although there is no guarantee that ρ will be positive, improvement in the expected reward under robust decisions has been documented empirically in the literature (Bertsimas et al. 2013, Kim and Lim 2014, Lorca and Sun 2014), and it will be seen in some of our examples. Regardless of its sign, $\frac{\rho}{2n}\delta$ is small relative to variance reduction and disappears when $n \to \infty$.

5.5. Summary

Figure 3 illustrates the key theoretical insights from this paper. Here, (A) is the expected reward $\mathbb{E}_{\mathbb{P}}[f(x, Y)]$ under the population distribution, (B1) is the distribution of the solution of the empirical problem $x_n(0)$, (B2) is the distribution of the out-of-sample reward $f(x_n(0), Y_{n+1})$, and (B3) is the "Jensen effect" for the empirical problem; (B1), (B2), and (B3) were previously shown in Figure 2. (C1), (C2), and (C3) are counterparts for the robust problem. Specifically, (C1) is the distribution of the robust solution, $x_n(\delta)$; (C2) is the distribution of the out-of-sample reward $f(x_n(\delta), Y)$ under the robust solution; and (C3) shows the Jensen effect $\mu^\star(\delta) - \mu_n(\delta)$. Indicated on (A) are the optimal rewards $\mu^\star(0)$ and $\mu^\star(\delta)$ of the population problems (see (12), (16), and (26)).

DRO makes a trade-off between expected reward and sensitivity. For the example in Figure 3, this corresponds to adding a bias to the empirical solution that pushes $x_n(0)$ toward the flat part of the reward function. This changes the distribution of the solution from (B1) to (C1) and the distribution of the out-of-sample reward from (B2) and (C2). In particular, pushing the solution toward the flat part of the objective reduces the variance of the out-of-sample reward (compare the spread of (B2) with that of (C2)) and changes the mean reward from $\mu_n(0)$ to $\mu_n(\delta)$. Theorem 3 quantifies the dependence of the mean and variance on δ.

Figure 3. Illustration of Key Theoretical Insights



Notes. (A) is the expected reward $\mathbb{E}_{\mathbb{P}}[f(x, Y)]$ under the population distribution. (B1), (B2), and (B3) are as described in Figure 2, and (C1), (C2), and (C3) are counterparts for the robust problem. Specifically, (C1) is the distribution of the robust solution, $x_n(\delta)$, (C2) is the distribution of the out-of-sample reward $f(x_n(\delta), Y)$ under the robust solution, and (C3) shows the Jensen effect $\mu^\star(\delta) - \mu_n(\delta)$. Indicated on (A) are the optimal rewards $\mu^\star(0)$ and $\mu^\star(\delta)$ of the population problems (see (12), (16), and (26)).

Robustness changes the Jensen effect from $\mu^\star(0) - \mu_n(0)$ for the SAA problem to $\mu^\star(\delta) - \mu_n(\delta)$ for the robust problem, indicated at (B3) and (C3). As indicated in the Equation (37) for ρ, the change in the magnitude of the Jensen effect is determined by the change in the variability of the solution and the change in the curvature of the reward, which can also be seen in this plot. For this example, the Jensen effect for the robust problem is smaller (i.e., $\rho > 0$), and robustness improves the out-of-sample expected reward ($\mu_n(\delta) > \mu_n(0)$) when δ is sufficiently small, which was discussed in Section 5.4. This is shown at (D).

Our analysis shows that relative to the SAA solution, substantial sensitivity reduction is possible with minimal impact on the expected reward when the robustness parameter (uncertainty set) is small. These results say nothing about this trade-off when δ is large, but the benefits of robustness in all of our experiments (see Section 7) are diminishing in δ. This suggests that the "small δ" regime is the one that is most interesting.

6. Calibration of Robust Optimization Models

Proposition 4 shows that when the ambiguity parameter δ is small, variance reduction is the first-order benefit of robust optimization, whereas the impact on the mean is (almost) an order of magnitude smaller. This implies that δ should be calibrated by trading off

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between the out-of-sample mean and variance and not just by optimizing the mean alone, as is commonly done when tuning the free parameters (e.g., in machine learning applications).

Observe, however, that the robust mean-variance frontier

$$\{(\mu_n(\delta),\, v_n(\delta)) : \delta \geq 0\} \equiv \{(\mathbb{E}_{\mathbb{P}}[f(x_n(\delta),\, Y_{n+1})],\ \mathbb{V}_{\mathbb{P}}[f(x_n(\delta),\, Y_{n+1})]) : \delta \geq 0\}$$

cannot be computed by the decision maker because he or she does not know the data generating model $\mathbb{P}$, so it is natural to approximate the frontier using resampling methods. One such approach uses the well-known *bootstrap* procedure (Efron and Tibshirani 1994), which we now describe and formally state in Algorithm 1.

Specifically, suppose the decision maker has a data set $\mathcal{D} = \{Y_1, \cdots, Y_n\}$ and the associated empirical distribution $\mathbb{P}_n$. To approximate the out-of-sample behavior of different possible data sets (of size n), one may generate a so-called *bootstrap data set*, calling it $\mathcal{D}^{(1)}$, by simulating n new i.i.d. data points from the empirical distribution $\mathbb{P}_n$. Associated with this bootstrap data set is the *bootstrap empirical distribution* $\mathbb{P}_n^{(1)}$ (Efron and Tibshirani 1994). This process can be repeated as many times as desired, with $\mathcal{D}^{(j)}$ and $\mathbb{P}_n^{(j)}$ respectively denoting the bootstrap data set and bootstrap empirical distribution generated at repeat j. We denote the number of bootstrap samples by k in Algorithm 1.

For each $\mathcal{D}^{(j)}$ and $\mathbb{P}_n^{(j)}$, we can compute (a family of) robust decisions $x^{(j)}(\delta)$ by solving the robust optimization problem defined in terms of the bootstrap empirical distribution $\mathbb{P}_n^{(j)}$ over a specified set of δ (step 4). The mean and variance of the reward for $x^{(j)}(\delta)$ under the *original empirical distribution* $\mathbb{P}_n$, which we denote by $\mu^{(j)}(\delta)$ and $v^{(j)}(\delta)$ in steps 5 and 6, can then be computed. The k bootstrap samples produces the mean-variance pairs $(\mu^{(j)}(\delta), v^{(j)}(\delta))$, $j = 1, \cdots, k$. Averaging these gives an estimate of the out-of-sample mean-variance frontier (steps 7 and 8).

Algorithm 1 (Bootstrap estimate of the out-of-sample robust mean-variance frontier generated by robust solutions)

Input: Data set $\mathcal{D} = \{Y_1, \ldots, Y_n\}$; ambiguity parameter grid $\mathcal{G} = \{\delta_1, \ldots, \delta_m\}$.

Output: Mean and variance of out-of-sample reward parametrized by $\delta \in \mathcal{G}$.

1 for $j \leftarrow 1$ **to** k **do**
2 $\quad \mathcal{D}^{(j)} \leftarrow$ bootstrap data set (sample n i.i.d. data points from $\mathbb{P}_n$);
3 $\quad$ for $i \leftarrow 1$ **to** m **do**
4 $\quad\quad x^{(j)}(\delta_i) \leftarrow \operatorname{argmax}_x \min_{\mathbb{Q}} \left\{\mathbb{E}_{\mathbb{Q}}[f(x,Y)] + \frac{1}{\delta_i}\mathcal{H}_\phi\left(\mathbb{Q}|\mathbb{P}_n^{(j)}\right)\right\}$,
5 $\quad\quad \mu^{(j)}(\delta_i) \leftarrow \mathbb{E}_{\mathbb{P}_n}[f(x^{(j)}(\delta_i), Y)]$,
6 $\quad\quad v^{(j)}(\delta_i) \leftarrow \mathbb{V}_{\mathbb{P}_n}[f(x^{(j)}(\delta_i), Y)]$.
7 $\hat{\mu}_n(\delta_i) \leftarrow \frac{1}{k}\sum_{j=1}^{k} \mu^{(j)}(\delta_i)$, for all $\delta_i \in \mathcal{G}$;
8 $\hat{v}_n(\delta_i) \leftarrow \frac{1}{k}\sum_{j=1}^{k} v^{(j)}(\delta_i) + \frac{1}{k-1}\sum_{j=1}^{k}(\mu^{(j)}(\delta_i) - \hat{\mu}_n(\delta_i))^2$, for all $\delta_i \in \mathcal{G}$;
9 **return** $\{(\hat{\mu}_n(\delta_i), \hat{v}_n(\delta_i)) : i = 1, \ldots, m\}$;

In the next section, we consider three applications: inventory control, portfolio optimization, and logistic regression. We illustrate various aspects of our theory and show how the bootstrap robust mean-variance frontier given in Algorithm 1 can be used to effectively calibrate ambiguity parameters in such settings.

7. Examples

We consider three examples. The first is a simulation experiment of the robust inventory control problem that illustrates key elements of our theory, and the next two examples are out-of-sample tests with real data. For these experiments, the penalty function $\mathcal{H}_\phi$ is relative entropy.

7.1. Example 1: Inventory Control

We first consider a simulation example with reward

$$f(x, Y) = r\min\{x, Y\} - cx. \tag{50}$$

This is a so-called inventory problem where x is the order quantity (decision), Y is the random demand, and r and c are the revenue and cost parameters, respectively.

The demand distribution $\mathbb{P}$ is a mixture of two exponential distributions, $\text{Exp}(\lambda_L)$ and $\text{Exp}(\lambda_H)$, where λ_L and λ_H are the rate parameters. This may correspond to two demand regimes (high and low) with different demand characteristics. For this numerical example, we set the mean values as $\lambda_L^{-1} = 10$ and $\lambda_H^{-1} = 100$ and revenue and cost parameters as $r = 30$ and $c = 2$, respectively. The probability that demand is drawn from the low segment is 0.7 (or equivalently, the probability that demand is drawn from the high segment is 0.3).

We run the following experiment. The decision maker is initially shown n data points $Y_1, \ldots, Y_n$ drawn i.i.d. from the mixture distribution $\mathbb{P}$. The decision maker then optimizes the robust objective function under the empirical distribution $\mathbb{P}_n$ to produce the optimal robust order quantity $x_n^\star(\delta)$. Another data point Y_{n+1} is then generated from $\mathbb{P}$, independent of the previous data points, and the objective value $f(x_n^\star(\delta), Y_{n+1})$ is recorded. The out-of-sample mean and variance $\mathbb{E}_{\mathbb{P}}[f(x_n^\star(\delta), Y_{n+1})]$ and $\mathbb{V}_{\mathbb{P}}[f(x_n^\star(\delta), Y_{n+1})]$ are approximated by running the experiment K times, where K is some large number, each time using a newly generated data set $Y_1, \ldots, Y_n, Y_{n+1} \sim \mathbb{P}$; the

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sample mean and variance are computed over the K repeats.

In Figure 4, we plot the pair $(\mathbb{E}_{\mathbb{P}}[f(x_n^\star(\delta), Y_{n+1})], \mathbb{V}_{\mathbb{P}}[f(x_n^\star(\delta), Y_{n+1})])$ for different sample sizes $n = 10, 30,$ and 50, with the marks on a given line corresponding to different values of δ; the rightmost mark corresponds to $\delta = 0$ (empirical). We also plot the "true" robust mean-variance frontier—that is, the pair $(\mathbb{E}_{\mathbb{P}}[f(x^\star(\delta), Y)], \mathbb{V}_{\mathbb{P}}[f(x^\star(\delta), Y)])$, which is independent of n.

Consistent with Theorem 3, we observe significant out-of-sample sensitivity (variance) reduction when δ is small with minimal impact on the out-of-sample expected reward. Our theory says nothing about the impact of large values of δ, though we see that the benefit of robustness is diminishing as δ increases, with the rate of sensitivity reduction diminishing and the expected reward being reduced at an increasing rate, as δ increases.

Figure 4 shows that as the sample size n increases, the gap between the out-of-sample mean-variance frontiers and the "true" robust mean-variance frontier gets smaller. This gap can be explained by Proposition 4, which shows that the difference between these frontiers should go to 0 like $O(n^{-1})$. This is shown in Figure 5, which plots the maximum gap (over δ) between the out-of-sample frontier and the true frontier under the true frontier for different values of n.

In the previous section, we proposed the bootstrap frontier as an approximation to the out-of-sample frontier. We next investigate the quality of this approximation for various sample sizes n. From the perspective of calibration, it is not necessary for the bootstrap frontier to equal the out-of-sample frontier; it need only preserve the *relative shape*. For example, if the only difference between the bootstrap frontier and the out-of-sample frontier was that it was double its size (in both the mean and variance), the choice of δ should be the same when using either frontier because the relative trade-off between mean and variance is identical. In light of this observation, in Figure 6, (a), (b), and (c), we plot the *normalized* bootstrap and out-of-sample frontiers, where the change in the mean and variance has been normalized to 1, for various sample sizes $n = 10, 30,$ and 50, respectively. It is clear that as the sample size increases, the bootstrap frontier more closely approximates the relative shape of the out-of-sample frontier.

Figure 4. Out-of-Sample Robust Mean-Variance Frontiers for $n = 10, 30,$ and 50 Data Points, and the True Frontier Generated by Solutions of DRO Under the Data Generating Model for Different Values of the Robustness Parameter δ

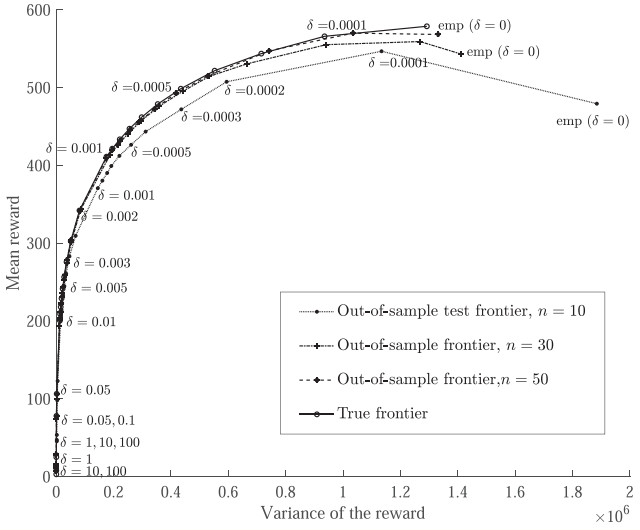
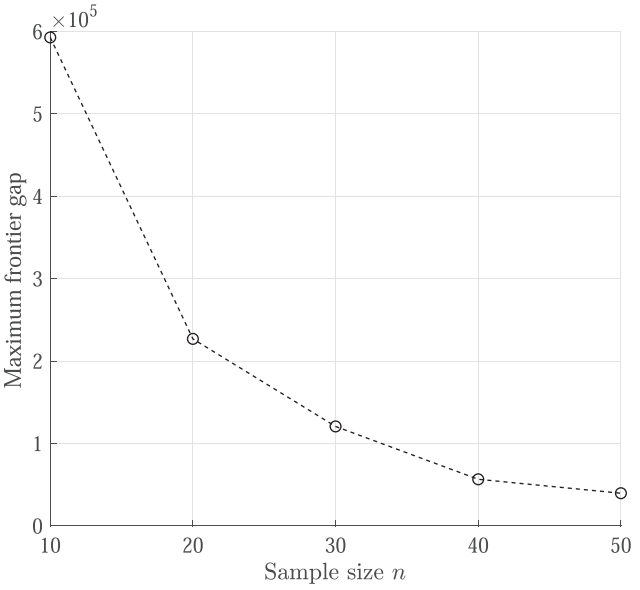


Figure 5. Gap Between the Out-of-Sample Frontier and True Frontier as n Gets Large



7.2. Example 2: Portfolio Optimization

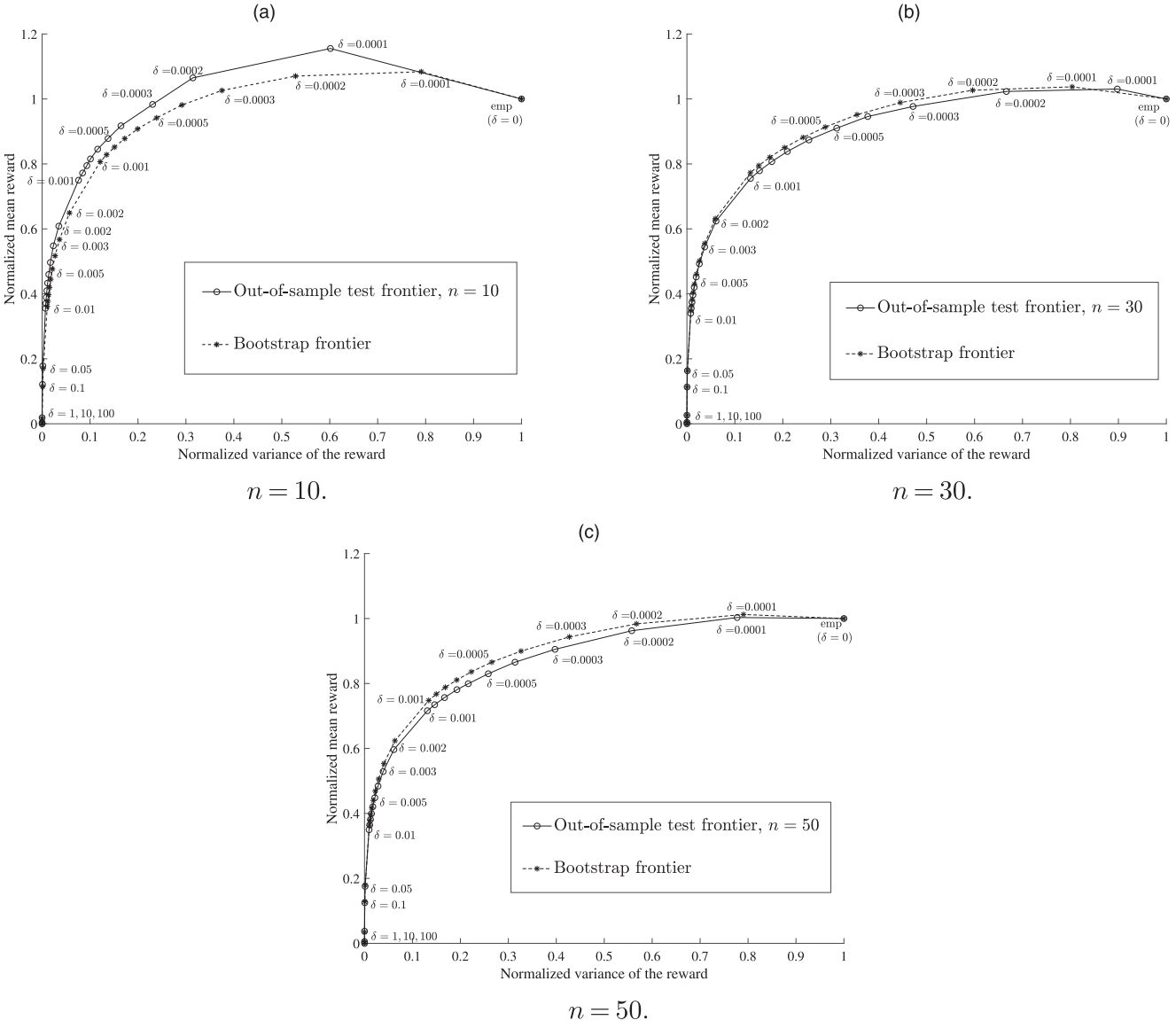
In our second application, we consider real monthly return data from the "10 Industry Portfolios" data set of French (2015). The reward function is exponential utility of returns,

$$f(x, R) = -\exp(-\gamma R' x), \tag{51}$$

where $x \in \mathbb{R}^d$ is the portfolio vector (decision variables), $R \in \mathbb{R}^d$ is the vector of random returns, and γ is the risk-aversion parameter. To simplify the experiments, we choose a risk-aversion parameter $\gamma = 1$. For the purposes of this example, we impose a budget constraint $1'x = 1$ and assume that asset holdings are bounded: $-1 \le x_i \le 1$, $i = 1, \ldots, d$.

We conduct the following experiment. We have $d = 10$ assets and are interested to see how robust

Figure 6. Bootstrap Frontier vs. Out-of-Sample Frontier (with Normalization)



Notes. Both frontiers are scaled and normalized so that both the mean and variance equal 1 when $\delta = 0$ (i.e., empirical optimization) and 0 in the most robust case ($\delta = 100$). We see that as n increases, the points on the frontier corresponding to the same values of δ converge.

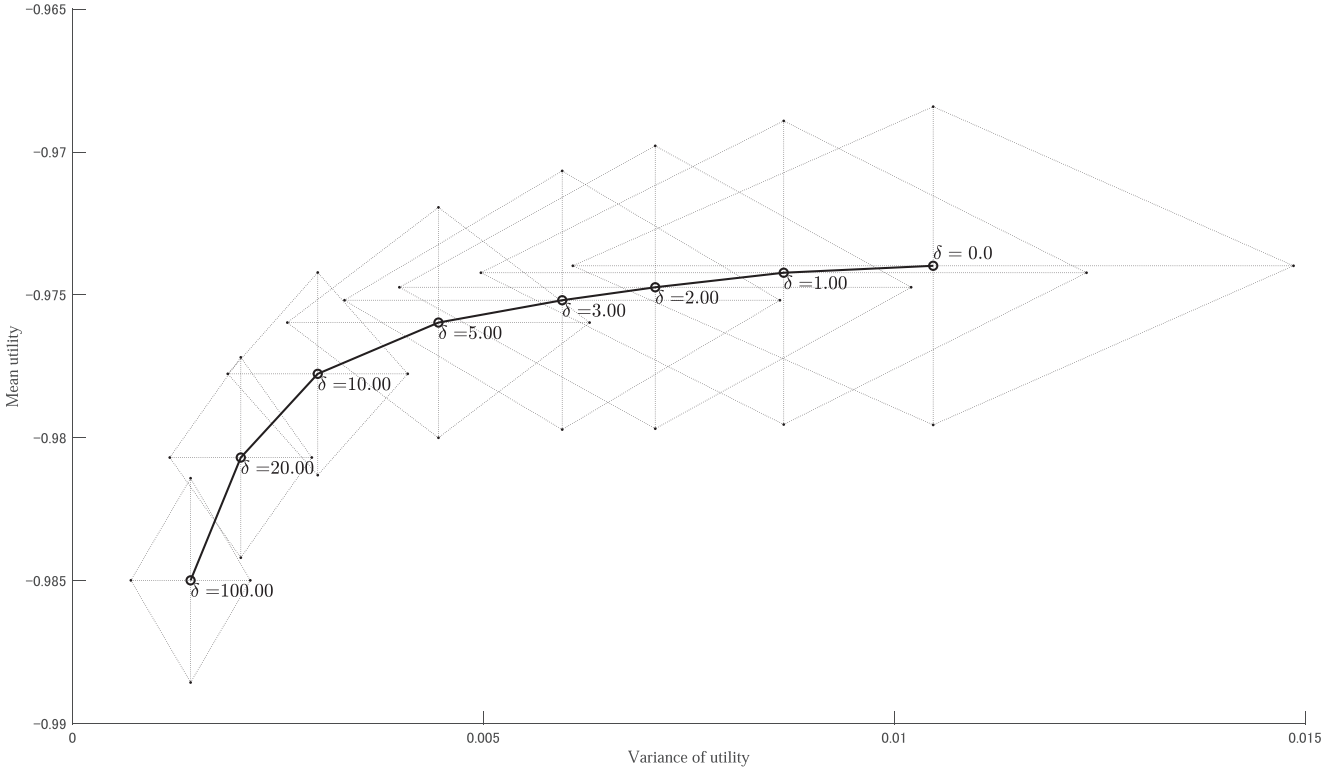
optimization and our approach for calibrating δ perform when we estimate the 10-dimensional joint distribution with relatively few data points ($n = 50$, for the time period from April 1968 to June 1972). The robust portfolios will be tested on the empirical distributions for monthly returns of future 50-month windows—July 1972 to September 1976, October 1976 to December 1980, and January 1981 to March 1985—that do not overlap with the training set.

To begin, we solve the robust portfolio choice problem using the 50 monthly returns for the period April 1968 to June 1972 for different values of δ and construct the robust mean-variance frontier using the bootstrap procedure described in Algorithm 1. Figure 7 shows this frontier, around which we also mark the ±1 standard deviations of the bootstrap samples in both the mean and variance dimensions. Empirical optimization corresponds to the point $\delta = 0$, and as predicted by Theorem 3, there is significantly more reduction in the variance compared with the mean when δ is close to 0. Our theory does not apply when δ is large. However, it shows that significant out-of-sample sensitivity reduction with minimal impact on the mean should be expected when δ is small, whereas Figure 7 suggests that the benefits of robustness are diminishing in δ, with the rate of sensitivity (variance) reduction decreasing and the rate of mean reduction increasing as δ increases.[2]

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Figure 7. Bootstrap Robust Mean-Variance Frontier for Portfolio Optimization Generated Using 50 Months of Monthly Return Data Between April 1968 and June 1972



7.2.1. Calibration and Out-of-Sample Tests. The bootstrap frontier estimates the out-of-sample mean and variance for different decisions and can be used to calibrate δ. For this example, the rate of variance reduction relative to the loss in the mean is substantial for $\delta \leq 5$, but this begins to diminish (and the cost of robustness increases) once δ exceeds 5. A value of δ between 2 and 5 seems reasonable. Although values of $\delta > 10$ may be preferred by some, the balance clearly tips toward loss in mean reward relative to variance reduction/robustness improvement.

Note also that a classical approach to calibration that optimizes the bootstrap estimate of the expected reward and ignores objective variability would select $\delta = 0$. This corresponds to empirical optimization and completely nullifies all the benefits of the robust model. More generally, although it is possible for DRO to produce solutions that outperform empirical optimization in terms of the out-of-sample expected reward, as proven in Theorem 3, such improvements are, in general, small relative to that of the variance reduction (when δ is small), and such solutions cannot be guaranteed even when model uncertainty is substantial, as shown by the portfolio choice example.

It is also interesting to compare calibration using the bootstrap frontier with the decisions obtained by solving the robust optimization problem

$$\max_{x} \min_{\mathbb{Q}\in\mathcal{U}_\alpha} \{\mathbb{E}_{\mathbb{Q}}[f(x,Y)]\}, \tag{52}$$

where the threshold $Q_n(\alpha)$ in the uncertainty set

$$\mathcal{U}_\alpha = \{\mathbb{Q} : \mathcal{H}(\mathbb{Q}\,|\,\mathbb{P}_n) \leq Q_n(\alpha)\} \tag{53}$$

is the $(1-\alpha)$-quantile of the distribution of $\mathcal{H}(\mathbb{Q}\,|\,\mathbb{P}_n)$ that we generate by simulating distributions $\mathbb{Q}$ by bootstrapping from $\mathbb{P}_n$. Confidence levels $1-\alpha$ with a value of 90%, 95%, or 99% are commonly suggested in the robust optimization literature.

It follows from duality that for any threshold $Q_n(\alpha)$, there is a unique ambiguity parameter value $\delta = \delta_\alpha > 0$ for which the solution of (52) coincides with our robust solution $x(\delta_\alpha)$; see corollary 3 in Ben-Tal et al. (2013) and also Online Appendix Section EC.7. In Table 1 we report the ambiguity parameter values δ_α for significance levels α corresponding to different points on the robust mean-variance frontier in Figure 7.

Although robust decisions associated with the "classical" significance levels of $\alpha = 0.01, 0.05,$ and 0.1 in Table 1 may or may not perform well out of sample in

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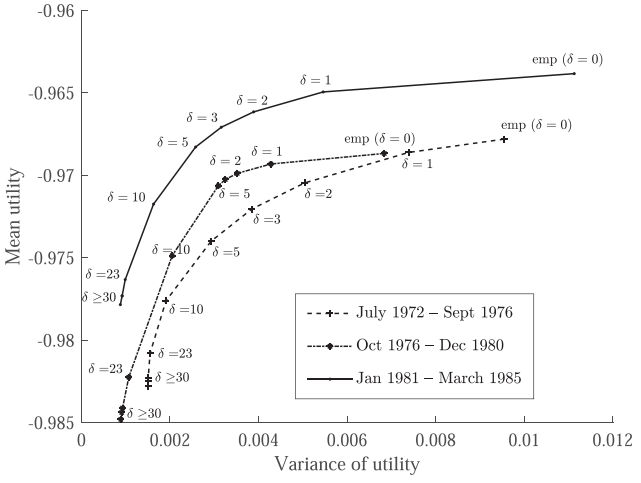
Table 1. Corresponding Ambiguity Parameter Values δ_α for Various Traditional Values of the Significance Level α

Significance level α	Ambiguity values δ_α
≈ 1	<10
0.8	23
0.10	30
0.05	31
0.01	33

any given application, it is clear that the range of δ_α associated with these significance levels is limited. This directly impacts the range of possible solutions available to the decision maker, which in this example are concentrated on the "extremely conservative" region of the bootstrap frontier (Figure 7). The values of $\delta \le 10$ associated with the desirable part of the frontier in Figure 7 correspond to significance levels α very close to 1. These significance levels are difficult to estimate using simulation because the associated quantiles of $\mathcal{H}(\mathbb{Q}|\mathbb{P}_n)$ are very close to 0. Although the asymptotic properties of $\mathcal{H}(\mathbb{Q}|\mathbb{P}_n)$ can be used to estimate α when the support of the data generating distribution $\mathbb{P}$ is finite and $\phi(z)$ is sufficiently regular, this seems pointless if we treat α as a parametrization of the trade-off between mean and sensitivity and accept that there is nothing special about classical significance levels when it comes to out-of-sample performance.

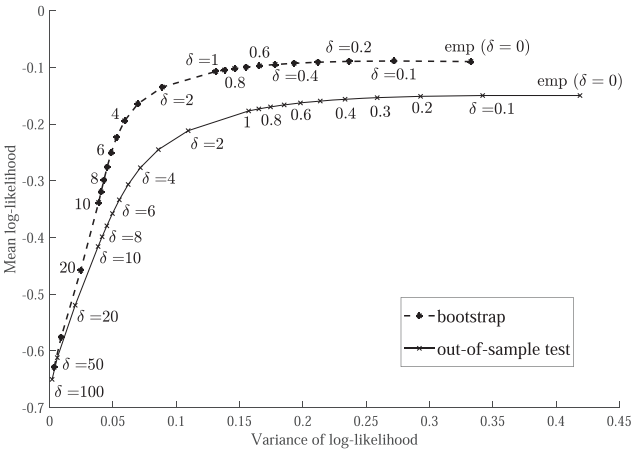
We next analyze the out-of-sample performance of the robust solutions obtained by solving the in-sample problem with $\delta = 1, 2, 3, 5,$ or 10 (as suggested by the bootstrap frontier in Figure 7); the solutions of (52) corresponding to $\delta_\alpha = 30$ ($\alpha = 0.10$), $\delta_\alpha = 31$ ($\alpha = 0.05$), and $\delta_\alpha = 33$ ($\alpha = 0.01$); and the solution of the empirical optimization problem (i.e., the robust solution with $\delta = 0$). We test each of the solutions on three out-of-sample test sets of data size $n = 50$ that do not overlap with the training data. Out-of-sample results are shown in Figure 8. Note that in the figure we report the mean and variance of the utility function $f(x, R)$ defined in (51), which is consistent with the framework of the robust optimization model in this paper.

Figure 8. Three Out-of-Sample Robust Mean-Variance Frontiers for the Portfolio Choice Problem



Notes. The frontiers are the average mean and variance over test data sets of 50 months.

Figure 9. Bootstrap Frontier vs. Out-of-Sample Test Frontier with WDBC Breast Cancer Diagnosis Data Set



Notes. The out-of-sample test frontier shows the mean and variance of the log-likelihood of the second half of the samples of the WDBC data set (i.e., 284 samples) on the basis of solutions obtained by using the first half (i.e., 285 samples). Here, we used three attributes—numbers 2, 24, and 25 (out of 30 available covariates)—which are found to be the best possible predictors in the paper by Mangasarian et al. (1995). To solve the optimization problems, we used RNUOPT (NTT DATA Mathematical Systems Inc.), a nonlinear optimization solver package.

Figure 8 consistently shows that as the value of δ ("robustness") increases from 0, the mean performance degrades but the objective variability is reduced, which is expected from our theory. Consistent with the bootstrap frontier (Figure 7), the portfolios associated with "high confidence uncertainty sets" have low variance, though the impact on expected utility is substantial.

7.3. Example 3: Logistic Regression

As a final application, we apply robust optimization to logistic regression, which we evaluate on the WDBC breast cancer diagnosis data set (Lichman 2013).

The reward function for logistic regression is given by

$$f((x, x_0), (Y, Y_0)) = -\ln\bigl(1 + \exp(-Y_0(x'Y + x_0))\bigr), \quad (54)$$

where $Y_0 \in \{-1, 1\}$ is the binary label; Y is the vector of covariates; and x and x_0 are decision variables representing coefficients and intercept, respectively, of the linear model for classification. Ordinary logistic regression is formulated as the maximization of the sample average of (54).

To demonstrate the out-of-sample behavior of robust maximum likelihood, we solve the ordinary/

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robust likelihood maximization problem using the first half of the WDBC breast cancer diagnosis data set (Lichman 2013) (i.e., 285 out of the 569 samples) and compute the log-likelihood and the variance of the log-likelihood of the resulting model using the remaining half of the samples (i.e., 284 out of 569). Figure 9 shows both the bootstrap frontier and the frontier obtained from the out-of-sample test. Once again, choices of δ that deliver good out-of-sample log-likelihood can be obtained using the bootstrap estimate of the robust frontier, with small values of δ delivering significant sensitivity (variance) reduction with minimal impact on the expected reward. As in the case of Figure 7, the benefits of robustness diminish as δ increases.

8. Conclusions

Proper calibration of DRO models requires a principled understanding of how the distribution of the out-of-sample reward depends on the "robustness parameter." In this paper, we studied out-of-sample properties of robust empirical optimization and developed a theory for data-driven calibration of the robustness parameter.

Robustness is closely related to controlling the variance of the reward distribution, as this reduces sensitivity of the mean reward to misspecification. Our main results show that the first-order benefit of "little bit of robustness" is a significant reduction in the variance of the out-of-sample reward, whereas the impact on the mean is almost an order of magnitude smaller. Our results imply that the robustness parameter should be calibrated by making trade-offs between estimates of the out-of-sample mean and variance.

To calibrate the robustness parameter, we introduced the robust mean-variance frontier and showed that it can be approximated using resampling methods such as the bootstrap. We applied the robust mean-variance frontiers to three applications: inventory control, portfolio optimization, and logistic regression. Our results showed that classical calibration methods that match "standard" confidence levels (e.g., 95%) are typically associated with an excessively large robustness parameter and overly pessimistic solutions that perform poorly out of sample, whereas ignoring the variance and calibrating purely on the basis of the mean can lead to a robustness parameter that is too small and a solution that misses out on the first-order benefits of robust optimization.

Endnotes

[1] We have added a scaling constant $-\frac{\phi''(1)}{\delta}$ in the second equation. This does not affect the solution of the first-order conditions but makes the subsequent analysis more convenient.

[2] Although our results apply in the regime when δ is small, we believe that similar methods can be used to derive expansions of the solution and out-of-sample reward in orders of $\eta \equiv 1/\delta$ around the solution of the DRO problem when $\eta = 0$ ($\delta = \infty$) and to study the out-of-sample performance of DRO solutions in this regime. Our experiments suggest that this is the less interesting part of the robust mean-variance frontier.

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Jun-ya Gotoh is a professor in the Department of Industrial and Systems Engineering at Chuo University. His research interests are in theory and applications of continuous optimization, especially for decision making under uncertainty, financial engineering, and data analysis.

Michael Jong Kim is an assistant professor in the Operations and Logistics Division, Sauder School of Business, at the University of British Columbia. His research interests are in sequential decision making, statistical learning, and robust optimization.

Andrew Lim is a professor in the Department of Analytics and Operations, the Department of Finance, and the Institute for Operations Research and Analytics at the National University of Singapore. His research interests are in the areas of stochastic control and optimization, decision making under uncertainty, robust optimization, and financial engineering. He is a past recipient of the National Science Foundation CAREER Award.

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